

EIDGENÖSSISCHE TECHNISCHE HOCHSCHULE ZÜRICH

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# Analysis I: One Variable

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Lecture Notes 2023-2024

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## Preface

Welcome to ETH Zürich and to your study of these lecture notes! A German version of this script was first produced in the academic year 2016/2017 by Manfred Einsiedler and Andreas Wieser for the lecture cycle Analysis I and II of the Interdisciplinary Natural Sciences Bachelor, Physics Bachelor, and Mathematics Bachelor programs. For the academic year 2019/2020, the lecture notes were heavily revised by Peter Jossen.

The current English version has been put together by Alessio Figalli for the academic year 2023/2024. Compared to the German version, some material has been reorganized. In addition, this version is more compact than its German correspondent and contains only the material that has been covered in class.

The Analysis I/II and Linear Algebra I/II lectures together form the foundation of mathematics as taught at ETH and also at other universities. The vast majority of future mathematics and physics lectures will build on these four lectures and extend their subject areas.

As we will see, the Analysis I/II lectures contain many topics related to differential and integral calculus. Some of these topics you probably already know from high school; nevertheless, we will not assume much prior knowledge apart from an intuitive understanding of variables and some practice for algebraic transformations. You will also notice an essential difference from high school teaching: although we will try to cover as many topics as possible with examples, these will be much less central than you are probably used to. Likewise, practicing the algorithms and calculation methods you have learned is not the focus of our lecture and is left to you to do on your own. Rather, the focus is on building the mathematical theory necessary to reach and understand topics such as differential equations and multidimensional integral theorems.

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# Chapter 1

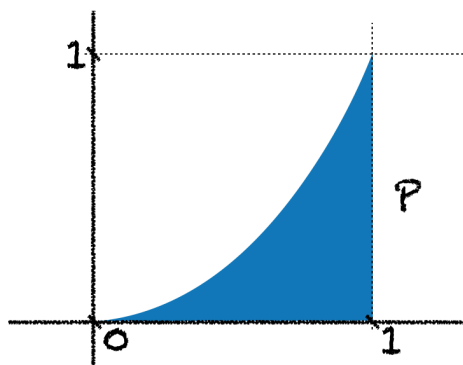
## Introduction

### 1.1 Quadrature of the Parabola

Before we begin our journey in the world of mathematical analysis, as an example of how we want to think and proceed here but also as an introduction to integral calculus, consider the set

$$P = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq x^2\}. \quad (1.1)$$

Our goal is to calculate its area.



This area was already determined as the first curvilinear bounded area by Archimedes (ca. 287–ca. 212 BCE) in the 3rd century BCE. For the area calculation, let us assume that we know what the symbols in the definition in equation (1.1) mean and that  $P$  describes the area in the following figure. In particular, we assume for the moment that we already know the set of real numbers  $\mathbb{R}$ .

Of course, calculating the area of  $P$  is not a challenge if we use integrals and the associated calculation rules. However, we do not want to assume we know the integral calculus. Strictly speaking, we must ask ourselves the following fundamental question before calculating:

*What is an area?*

If we cannot answer this question exactly, then we cannot know what it means to calculate the area of  $P$ . Therefore, we qualify our goal in the following way:

PROPOSITION 1.1. — Suppose there is a notion of area in  $\mathbb{R}^2$  that satisfies the following properties:

1. The area of the rectangle  $[a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$  is equal to  $(b - a)(d - c)$ , where  $a, b, c, d$  are real numbers with  $a \leq b, c \leq d$ .
2. If  $G$  is a domain in  $\mathbb{R}^2$  and  $F$  is a domain contained in  $G$ , then the area of  $F$  is less than or equal to the area of  $G$ .
3. For sets  $F, G$  in  $\mathbb{R}^2$  without common points, the area of the union  $F \cup G$  is the sum of the areas of  $F$  and  $G$ .

Then the area of  $P$  as in equation (1.1) (if defined at all) is equal to  $\frac{1}{3}$ .

In other words, we have left open the question of whether there is a notion of area and for what areas it is defined, but we want to show that  $\frac{1}{3}$  is the only reasonable value for the area of  $P$ .

For the proof of Proposition 1.1 we need a lemma (also called an “auxiliary theorem”):

LEMMA 1.2. — Let  $n \geq 1$  be a natural number. Then

$$1^2 + 2^2 + \cdots + (n-1)^2 + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}. \quad (1.2)$$

1

*Proof.* We perform the proof using induction. For  $n = 1$ , the left-hand side of equation (1.2) is equal to 1 and the right-hand side is equal to  $\frac{1}{3} + \frac{1}{2} + \frac{1}{6} = 1$ . So equation (1.2) is true for  $n = 1$ . This part of the proof is called the beginning of induction.

Suppose we already know that equation (1.2) holds for the natural number  $n$ . We now want to show that it follows that equation (1.2) also holds for  $n + 1$ . The left-hand side of equation (1.2), for  $(n + 1)$  instead of  $n$ , is given by

$$1^2 + 2^2 + \cdots + n^2 + (n + 1)^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} + (n + 1)^2 = \frac{n^3}{3} + \frac{3n^2}{2} + \frac{13n}{6} + 1$$

where, in the first equality, we have used the validity of equation (1.2) for the number  $n$ . The right-hand side of equation (1.2), for  $(n + 1)$  instead of  $n$ , is given by

$$\begin{aligned} \frac{(n+1)^3}{3} + \frac{(n+1)^2}{2} + \frac{n+1}{6} &= \frac{n^3 + 3n^2 + 3n + 1}{3} + \frac{n^2 + 2n + 1}{2} + \frac{n+1}{6} \\ &= \frac{n^3}{3} + \frac{3n^2}{2} + \frac{13n}{6} + 1 \end{aligned}$$

This shows that the left and right sides of equation (1.2) also agree for  $n + 1$ . This part of the proof is called the induction step.

It follows that equation (1.2) is true for  $n = 1$  due to the validity of the beginning of the induction. Therefore, it is also true for  $n = 2$  due to the induction step, and for  $n = 3$  again due to the induction step. Continuing in this way, we obtain (1.2) for any natural number.



We say that equation (1.2) follows by means of induction for all natural numbers  $n \geq 1$ . Furthermore, we indicate the end of the proof with a small square.  $\square$

*Proof of Proposition 1.1.* We assume that there is a notion of area content with the properties in the proposition and that it is defined for  $P$ . Suppose that  $I$  is the area content of  $P$ . We cover  $P$  for a given natural number  $n \geq 1$  with rectangles whose base has length  $\frac{1}{n}$ , as in Figure 1.1 on the left.

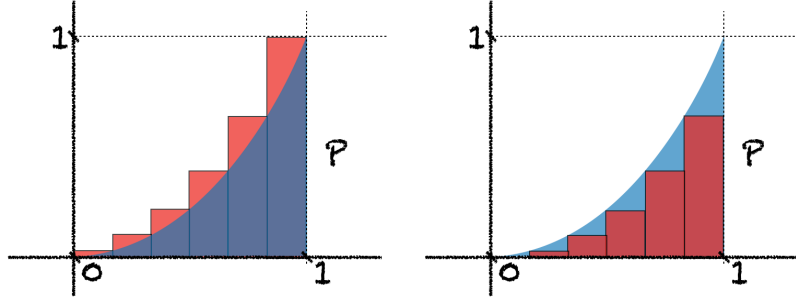


Figure 1.1: Approximation of  $P$  with  $n = 6$  rectangles.

Note that the first rectangle from the left has height  $(\frac{1}{n})^2$ , the second one  $(\frac{2}{n})^2$ , and so on. Hence, from the assumed properties of the area, thanks to Lemma 1.2 the following inequality holds:

$$\begin{aligned} I &\leq \frac{1}{n} \frac{1}{n^2} + \frac{1}{n} \frac{2^2}{n^2} + \cdots + \frac{1}{n} \frac{n^2}{n^2} \\ &= \frac{1}{n^3} (1^2 + 2^2 + \cdots + n^2) \\ &= \frac{1}{n^3} \left( \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} \right) \\ &= \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}. \end{aligned}$$

Note that the straight-line segments where the rectangles touch have area 0, and we may ignore them.

If, on the other hand, we use rectangles as in Figure 1.1 on the right, we also get

$$\begin{aligned} I &\geq \frac{1}{n} \frac{0}{n^2} + \frac{1}{n} \frac{1^2}{n^2} + \cdots + \frac{1}{n} \frac{(n-1)^2}{n^2} \\ &= \frac{1}{n^3} (1^2 + \cdots + (n-1)^2) \\ &= \frac{1}{n^3} (1^2 + \cdots + (n-1)^2 + n^2 - n^2) \\ &= \frac{1}{n^3} \left( \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} - n^2 \right) \\ &= \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2} \end{aligned}$$

So in summary

$$-\frac{1}{2n} + \frac{1}{6n^2} \leq I - \frac{1}{3} \leq \frac{1}{2n} + \frac{1}{6n^2} \quad (1.3)$$

for all natural numbers  $n \geq 1$ . The only number that satisfies this for all natural numbers  $n \geq 1$  is 0. Therefore,  $I - \frac{1}{3} = 0$  and the proposition follows.  $\square$

To rigorously prove the statement above, one needs to show that the only real number satisfying (1.3) for all  $n \geq 1$  is 0. This is intuitively clear: indeed, taking  $n$  larger and larger, the two expressions  $\frac{1}{2n} + \frac{1}{6n^2}$  and  $-\frac{1}{2n} + \frac{1}{6n^2}$  get smaller and smaller. However, we cannot give a proof of it at this point, as we lack a rigorous definition of the real numbers.

As already mentioned, we have not answered the question of whether there is a notion of area for sets in  $\mathbb{R}^2$ . Nor have we described precisely what domains in  $\mathbb{R}^2$  are, but we have implicitly assumed that domains are those subsets of  $\mathbb{R}^2$  to which we can assign an area. The notions of the Riemann and Lebesgue integrals and measurable sets answer these fundamental questions.

The idea of the proof is illustrated in the following applet.

**Applet 1.3** ([Estimating an area](#)). *We use up to 1000 rectangles to estimate the area from below and from above. In the proof below, however, we will use an unlimited number of rectangles and can thus determine the area exactly without any fuzziness.*

1

Note that in the previous examples, we informally used the notion of “set”. Here, for completeness, we give a more precise definition.

#### INTERLUDE: NAIVE SET THEORY

The central assumptions of naive set theory are the following postulates (1), (2), and (4). Postulate (3), which we state here in addition, does not usually belong to them, but is a consequence of the so-called *axiom of regularity* in Zermelo-Frenkel set theory.

- (1) A **set** consists of distinguishable elements.
- (2) A set is distinctively determined by its **elements**.
- (3) A set is not an element of itself.
- (4) Every statement  $A$  about elements of a set  $X$  defines the set of elements in  $X$  for which the statement  $A$  is true; one writes  $\{x \in X \mid A \text{ is true for } x\}$ .

The **empty set**, written as  $\emptyset$  (or sometimes also  $\{\}$ ), is the set containing no elements.

We write “ $x \in X$ ” if  $x$  is an element of the set  $X$ . If  $x$  is not an element of the set  $X$ , we write  $x \notin X$ . Sometimes we describe a set by a concrete list of its elements, for example,  $X = \{x_1, x_2, \dots, x_n\}$ , but often it is more convenient to use postulate (4). For example, one

can write

$$\{n \in \mathbb{Z} \mid \exists m \in \mathbb{Z} : n = 2m\}$$

1

to describe the set of even numbers. The symbol  $\exists$  means “there exists”, while the symbol  $\forall$  means “for all”. The symbols “ $\mid$ ” and “ $:$ ” in the formula above both mean “such that”. With time, this mathematical terminology will become familiar, one just needs time and practice.

## 1.2 Tips on Studying

*“All beginnings are difficult”*

You will notice a big difference between school mathematics and university mathematics. The latter also uses its own language, which you will have to learn. The sooner you take this on, the more you will take away from the lectures. This brings us to the next tip.

*“There is no silver bullet to mathematics”*

(handed down quotation from Euclid to the Egyptian king Ptolemy I)

You cannot learn mathematics by watching it; just as you cannot learn tennis or skiing by watching all available tournaments or world championships on television. Rather, you should learn mathematics like a language, and a language is taught by using it. Discuss the topics of the lectures with colleagues. Explain to each other the proofs from the lecture or the solution of the exercise examples. Above all, solve as many exercises as possible; this is the only way to be sure that you have mastered the topics.

It is fine to work on the exercises in small groups. This even has the advantage that the group discussions make the objects of the lectures more lively. However, you should ensure that you fully understand the solutions, explain them and subsequently solve similar problems on your own.

*“He who asks is a fool for a minute. He who does not ask is a fool all his life.”*

Confucius

Ask as many questions as you can and then ask them when they come up. Probably many of your colleagues have the same question, or have not even noticed the problem. This allows the lecturer or teaching assistant to simultaneously fix a problem for many and identify problems in students where she or he thought none existed. Furthermore, good question formulation needs to be practiced; the first year is the ideal time to do this.

---

## Chapter 2

# The Real Numbers: Maximum, Supremum, and Sequences

### 2.1 The Axioms of the Real Numbers

#### 2.1.1 Ordered Fields

Although we all have an intuitive notion of real numbers, our aim is to introduce them properly. To do so, we shall first introduce the notion of a field and then the concept of a relation.

We start by introducing some fundamental notions that will be covered in much more depth in the algebra lectures. Loosely speaking, a group is a set equipped with an “operation” that satisfies a list of properties. For our purposes, it is enough to know that an operation is something that takes two elements of a set and gives back a third element of the set. We now specify which properties the operation has to satisfy so that we can speak of a group.

## INTERLUDE: GROUPS

A **group** is a (non-empty) set  $G$  endowed with an operation “ $\star$ ” that satisfy the following properties:

- (Associativity) for all  $a, b, c \in G$ , we have

$$(a \star b) \star c = a \star (b \star c).$$

- (Neutral element) there exists a neutral element, i.e.,  $e \in G$  such that for all  $a \in G$  we have

$$a \star e = e \star a = a.$$

- (Inverse element) for each element  $a \in G$  there is an inverse element, i.e.,  $a^{-1} \in G$  such that

$$a \star a^{-1} = a^{-1} \star a = e.$$

Note that, in general, we do not require that  $a \star b = b \star a$  for arbitrary  $a, b \in G$ . If this property holds, the group is called **commutative** or **abelian**.

2

EXAMPLE 2.1. — To make these concepts easier to access, let us assume for the moment that we already know the natural numbers  $\mathbb{N}$  and the integers  $\mathbb{Z}$ . We check whether they satisfy the above properties if we replace  $\star$  with the operations you are already familiar with.

1. Consider the natural numbers  $\mathbb{N} = \{0, 1, 2, 3, \dots\}$  with the usual addition  $+$  that you probably know since primary school. This is

- associative, since for any natural numbers  $k, l, m \in \mathbb{N}$  we have

$$(k + l) + m = k + (l + m),$$

- has a neutral element  $0 \in \mathbb{N}$ , since for any natural number  $n \in \mathbb{N}$  we have

$$0 + n = n + 0 = n,$$

but

- no element apart from  $0$  has an inverse element. In fact, the inverse element of  $n \in \mathbb{N} \setminus \{0\}$  would be  $-n$ , which is not included in the natural numbers  $\mathbb{N}$ .

2. The same arguments show that the integers  $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ , again with the addition, form a group. Moreover, this is a commutative group, since for all integers  $n, m \in \mathbb{Z}$  we have

$$n + m = m + n.$$

3. As a different example, consider the set of nonzero rational numbers

$$\mathbb{Q}^\times = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, p, q \neq 0 \right\}$$

with the usual multiplication  $\cdot$  between numbers. In this case, one can check that the multiplication is associative and commutative, the neutral element is 1, and the inverse of  $\frac{p}{q}$  is  $\frac{q}{p}$ . Hence, this is a commutative group.

2.2. — It follows directly from the definition of the neutral element that it is unique. Indeed, assume that additionally to  $e \in G$ , we have a second element  $e'$  with the property such that  $e' \star a = a \star e' = a$  for all elements  $a \in G$ . Then, we can choose  $a = e$  and obtain

$$e = e \star e' = e',$$

where the first equality follows from the fact that  $e'$  is neutral, while in the second equality we used that  $e$  is neutral.

We can thus speak of *the* neutral element of a group.

In the same spirit, assume that for an element  $a \in G$ , there exist two inverse elements  $a^{-1}$  and  $\tilde{a}^{-1}$ . Then, using associativity, we observe that

$$a^{-1} = a^{-1} \star e = a^{-1} \star (a \star \tilde{a}^{-1}) = (a^{-1} \star a) \star \tilde{a}^{-1} = e \star \tilde{a}^{-1} = \tilde{a}^{-1}.$$

So also for inverse elements, we might speak of *the* inverse element. In particular, since  $a \star a^{-1} = e$ , we deduce that  $a$  is the inverse of  $a^{-1}$ , thus

$$(a^{-1})^{-1} = a. \tag{2.1}$$

## INTERLUDE: RINGS AND FIELDS

A **ring** is a (non-empty) set  $R$  in which we can “add” and “multiply” elements in a compatible way. More precisely, a ring is a commutative group  $R$  whose first operation “+” is called “addition”, and it is also equipped with an additional operation called “multiplication” and denoted by “ $\cdot$ ”.

The neutral element for the addition is denoted by 0. The multiplication  $\cdot$  is associative, has a neutral element (usually denoted by 1), and satisfies the following property:

- (Distributivity) for all  $a, b, c \in R$ , we have

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

and

$$(a + b) \cdot c = a \cdot c + b \cdot c.$$

A ring in which the operation  $\cdot$  is also commutative is called a **commutative ring**. Note that, in a ring, we do not require that elements have an inverse for the multiplication.

A commutative ring in which every “non-zero” element (that is, any element other than the neutral element 0 for the addition) has an inverse element for the multiplication  $\cdot$  is called a **field**. In other words, a ring  $R$  is a field if  $R \setminus \{0\}$  is a commutative group for the operation  $\cdot$ .

Fields will usually be denoted by the letter  $K$  for the corresponding German word “Körper”. We write  $K^\times = K \setminus \{0\}$  for the set of invertible elements in  $K$ .

EXAMPLE 2.3. — Let us continue with our examples. We have already established that the integers  $\mathbb{Z}$  form a commutative group, but are they also a ring with the usual multiplication? We must check:

- Associativity of the multiplication: For all integers  $k, l, m \in \mathbb{Z}$ , we have

$$(k \cdot l) \cdot m = k \cdot (l \cdot m).$$

- Neutral element for the multiplication: The neutral element for the multiplication is  $1 \in \mathbb{Z}$  as, for all integers  $k \in \mathbb{Z}$ , we have

$$1 \cdot k = k \cdot 1 = k.$$

- Distributivity: For all  $k, l, m \in \mathbb{Z}$  we have

$$k \cdot (l + m) = k \cdot l + k \cdot m$$



and

$$(k + l) \cdot m = k \cdot m + l \cdot m.$$

Hence, we conclude that  $\mathbb{Z}$  is a ring. Moreover, since the multiplication is commutative (namely,  $k \cdot l = l \cdot k$ ), it is a commutative ring. However, it is not a field since no element other than 1 and  $-1$  has a multiplicative inverse. For example, the multiplicative inverse of 2 is  $\frac{1}{2}$ , which is not an integer.

EXAMPLE 2.4. — The set of rational numbers  $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$  with the usual addition and multiplication is a field.

2.5. — Before going on, we look at some immediate consequences of the definition of field. In the current notation,  $-a$  denotes the inverse of  $a$  with respect to the addition, while  $a^{-1}$  is the inverse of  $a$  with respect to the multiplication. Note that, in the current context, (2.1) implies that

$$-(-a) = a, \quad \text{and} \quad (a^{-1})^{-1} = a \quad \text{whenever} \quad a \neq 0. \quad (2.2)$$

Let  $K$  be a field, and  $a, b \in K$ . Then the following holds:

- (i)  $0 \cdot a = 0$  and  $a \cdot 0 = 0$ .

Proof: Since 0 is the neutral element for the addition, we have  $0 = 0 + 0$ . Hence, using distributivity, we get

$$0 \cdot a = (0 + 0) \cdot a = (0 \cdot a) + (0 \cdot a).$$

Adding  $-0 \cdot a$  (i.e., the inverse of  $0 \cdot a$  for the addition), we deduce that  $0 \cdot a = 0$ . The case of  $a \cdot 0$  is analogous.

- (ii)  $a \cdot (-b) = -(a \cdot b) = (-a) \cdot b$ . In particular, we have  $(-1) \cdot a = -a$ .

Proof: By the distributive law, we have

$$a \cdot b + a \cdot (-b) = a \cdot (b + (-b)) = a \cdot 0 = 0.$$

So  $a \cdot (-b)$  is the additive inverse of  $a \cdot b$ , i.e.,  $-(a \cdot b) = a \cdot (-b)$ . Taking  $b = 1$  gives  $-a = (-1) \cdot a$ .

The validity of  $(-a) \cdot b = -(a \cdot b)$  follows exchanging  $a$  and  $b$  in the argument above.

- (iii)  $(-a) \cdot (-b) = a \cdot b$ . In particular, we have  $(-a)^{-1} = -(a^{-1})$ .

Proof: By (ii) we know that  $-(a \cdot b) = a \cdot (-b)$ . Hence, recalling (2.2),

$$a \cdot b = -(a \cdot (-b)).$$

On the other hand, applying (ii) with  $(-b)$  instead of  $b$ , we also have

$$-(a \cdot (-b)) = (-a) \cdot (-b).$$

Combining the two identities above, we conclude that  $(-a) \cdot (-b) = a \cdot b$ . Finally, taking  $b = a^{-1}$  yields  $(-a) \cdot (-(a^{-1})) = a \cdot a^{-1} = 1$ , which gives the second assertion.

REMARK 2.6. — A natural question one may ask is the following: *Is it possible to construct a field  $K$  where  $0$  (i.e., the neutral element for  $+$ ) and  $1$  (i.e., the one for  $\cdot$ ) are equal?*

Assume that  $0 = 1$ . Then, using (i) above and the fact that  $1$  is the neutral element for multiplication, we get

$$0 = a \cdot 0 = a \cdot 1 = a$$

for every  $a \in K$ . So, the only possibility for having  $0 = 1$  is that  $K$  consists of the single element  $0$ . From now on, we shall assume that  $K$  always contains at least two elements, so in particular,  $0$  and  $1$  cannot coincide.

Next, we introduce the second ingredient of an ordered field, the order relation. Again we will do so in steps:

#### INTERLUDE: CARTESIAN PRODUCT

Let  $X$  and  $Y$  be two sets. The **cartesian product**  $X \times Y$  is the set of ordered pairs of elements in  $X$  and  $Y$ :

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}.$$

EXAMPLE 2.7. — The cartesian product  $X \times Y$  of  $X = \{A, B, C, D, E, F, G, H\}$  and  $Y = \{1, 2, 3, 4, 5, 6, 7, 8\}$  is what we use to write down positions on a chess board: To each pair we can associate a unique square on the chess board and vice versa. For instance, the black king starts the game on the square corresponding to  $(E, 8) \in X \times Y$ .

#### INTERLUDE: SUBSETS

Let  $P$  and  $Q$  be sets.

- We say that  $P$  is **subset** of  $Q$ , and write  $P \subset Q$  (or  $P \subseteq Q$ ), if for all  $x \in P$  also  $x \in Q$  holds.
- We say that  $P$  is a **proper subset** of  $Q$ , and write  $P \subsetneq Q$ , if  $P$  is a subset of  $Q$  but not equal to  $Q$ .
- We write  $P \not\subset Q$  (or  $P \not\subseteq Q$ ) if  $P$  is not a subset of  $Q$ .

Equivalent formulations for “ $P$  is a subset of  $Q$ ” are “ $P$  is **contained** in  $Q$ ” and “ $Q$  is a **supset** of  $P$ ”, which we also write as “ $Q \supset P$ ”. The meaning of the statement “ $Q$  is a **proper superset** of  $P$ ”, written  $Q \supsetneq P$ , is now implicit. Because of the second assumption of naive set theory, two sets  $P$  and  $Q$  are equal exactly if both  $P \subset Q$  and  $Q \subset P$  hold.

For instance,  $\{x, y\} = \{z\}$  holds if  $x = y = z$ . Note that there are no “multiplicities” for elements of a set (for instance,  $\{x, x, x\} = \{x\}$ ).

#### INTERLUDE: RELATIONS

Let  $X$  be a set. A **relationship** on  $X$  is a subset  $\mathcal{R} \subset X \times X$ , that is, a list of ordered pairs of elements of  $X$ . We also write  $x\mathcal{R}y$  if  $(x, y) \in \mathcal{R}$  and often use symbols such as  $<, \leq, \cong, \equiv, \sim$  for the relations.

If  $\sim$  is a relation, we write “ $x \not\sim y$ ” if “ $x \sim y$ ” does not hold. A relation  $\sim$  is called:

1. (Reflexive) for all  $x \in X : x \sim x$ .
2. (Transitive) for all  $x, y, z \in X : x \sim y$  and  $y \sim z \implies x \sim z$ .
3. (Symmetric) for all  $x, y \in X : x \sim y \implies y \sim x$ .
4. (Antisymmetric) for all  $x, y \in X : x \sim y$  and  $y \sim x \implies x = y$ .

A relation is called **equivalence relation** if it is reflexive, transitive, and symmetric.

A relation is called **order relation** if it is reflexive, transitive, and antisymmetric.

EXAMPLE 2.8. — We look again at the integers  $\mathbb{Z}$  and two examples of relations on them. Let  $m, n, p \in \mathbb{Z}$  be integers.

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- Consider the relation  $\leq$  of being “less than or equal to”, i.e., we write  $n \leq m$  if  $n$  is less than or equal to  $m$ . Then we see that this relation is:
  1. reflexive because of the trivial observation that  $n = n$ ,
  2. transitive since  $n \leq m$  and  $m \leq p$  implies  $n \leq p$ ,
  3. not symmetric since for instance  $7 \leq 8$  but  $8 \not\leq 7$ ,
  4. antisymmetric since if  $n$  is less than or equal to  $m$  and vice versa, the only possibility is that they are equal.

We conclude that  $\leq$  is an order relation.

- Consider next the relation  $<$  of being “strictly smaller than”, i.e., we write  $n < m$  if  $n$  and  $m$  are distinct integers and  $n$  is less than  $m$ . This relation is:
  1. not reflexive since no integer is strictly smaller than itself,
  2. transitive since  $n < m$  and  $m < p$  implies  $n < p$ ,
  3. not symmetric since  $3 < 5$  but  $5 \not< 3$ ,
  4. antisymmetric. This point is a bit subtle: We need to check for all  $n, m$  such that  $n < m$  and  $m < n$ , we have  $n = m$ . But there are no such  $n, m$  and hence the condition of antisymmetry is fulfilled because there is nothing to check.

We conclude that  $<$  is neither an equivalence relation nor an order relation, since it does not satisfy the reflexivity property.

#### DEFINITION 2.9: ORDERED FIELD

Let  $K$  be a field, and let  $\leq$  be an order relation on the set  $K$ . We call  $(K, \leq)$ , or  $K$  for short, an **ordered field** if the following conditions are satisfied:

1. (Linearity of the order) for all  $x, y \in K$ ,

at least one between  $x \leq y$  and  $y \leq x$  holds.

2. (Compatibility of order and addition) for all  $x, y, z \in K$ , it holds

$$x \leq y \implies x + z \leq y + z.$$

3. (Compatibility of order and multiplication) for all  $x, y \in K$ , it holds

$$0 \leq x \text{ and } 0 \leq y \implies 0 \leq x \cdot y.$$

The following terminology is standard and will be used throughout these lecture notes:

- We pronounce  $x \leq y$  as “ $x$  is less than or equal to  $y$ ”.
- For  $x, y \in K$  we define  $y \geq x$  by  $x \leq y$ , and pronounce this as “ $y$  is greater than or equal to  $x$ ”.
- We say that an element  $x \in K$  is non-negative if  $x \geq 0$ , and non-positive if  $x \leq 0$ .
- Further, we define  $x < y$  (pronounced as “ $x$  is smaller than  $y$ ” or “ $x$  is strictly smaller than  $y$ ”) whenever  $x \leq y$  and  $x \neq y$ .
- Analogously, we define  $x > y$  when  $y < x$ , and say “ $x$  is greater than  $y$ ” or “ $x$  is strictly greater than  $y$ ”.
- An element  $x \in K$  is positive if  $x > 0$  holds, and negative if  $x < 0$  holds.

We often use these symbols in “equidirectional chains”, for example,  $x \leq y < z = a$  stands for

$$x \leq y \quad \text{and} \quad y < z \quad \text{and} \quad z = a.$$

EXAMPLE 2.10. — A well-known example of an ordered field is the one of rational numbers  $\mathbb{Q}$ , together with the usual order relation given by

$$\frac{p}{q} \leq \frac{p'}{q'} \iff pq' \leq p'q, \quad p, p' \in \mathbb{Z}, q, q' \in \mathbb{N}.$$

Here on the right-hand side is the order on the integers, which we assume to be known.

2.11. — Let  $(K, \leq)$  be an ordered field, and  $x, y, z, w$  denote elements of  $K$ . We want to prove a series of properties that follow from the definitions. To simplify the notation, it is customary to write  $\cdot$  for multiplication only if it would otherwise be confusing. This is why, in proofs,  $\cdot$  may disappear. For example, we may write  $xy$  instead of  $x \cdot y$ .

- (a) (Trichotomy) *Either  $x < y$  or  $x = y$  or  $x > y$ .*

Proof: This follows directly from the linearity of the order relation  $\leq$ .

- (b) *If  $x < y$  and  $y \leq z$ , then  $x < z$  also holds.*

Proof: First of all, since our assumption implies in particular that  $x \leq y$  and  $y \leq z$ , we have  $x \leq z$  according to the transitivity of the order relation.

We now want to prove  $x < z$ . Assume by contradiction that this is not the case. Then, since  $x \leq z$ , it must be  $x = z$ . Recalling that  $y \leq z$ , this proves that  $y \leq x$ . However, this contradicts  $x < y$ . Hence, we conclude that  $x = z$  is impossible and, therefore,  $x < z$ .

Analogously, we see that  $x \leq y$  and  $y < z$  imply that  $x < z$ .

- (c) (Addition of Inequalities) *If  $x \leq y$  and  $z \leq w$  hold, then  $x + z \leq y + w$  also holds.*

Proof: Indeed,  $x \leq y$  implies  $x + z \leq y + z$  according to the additive compatibility in Definition 2.9, and  $z \leq w$  implies  $y + z \leq y + w$  for the same reason. Transitivity of the order relation implies  $x + z \leq y + w$ .

Analogously, using inference (b), one sees that  $x < y$  and  $z \leq w$  imply that  $y + z < y + w$ .

- (d)  *$x \leq y$  is equivalent to  $0 \leq y - x$  holds.*

Proof: If  $x \leq y$ , then adding  $-x$  to both sides, we obtain  $0 \leq y - x$ .

Vice versa, if  $0 \leq y - x$  then adding  $x$  to both sides we obtain  $x \leq y$ .

- (e)  *$x \leq 0$  is equivalent to  $0 \leq -x$ .*

Proof: This follows from (d) with  $y = 0$ .

- (f)  *$x^2 \geq 0$ , and  $x^2 > 0$  if  $x \neq 0$ .*

Proof: If  $x \geq 0$ , the first statement follows from the multiplicative compatibility in Definition 2.9.

If  $x \leq 0$ , then  $-x \geq 0$  by inference (e) and hence  $x^2 = (-x)^2 \geq 0$  (the fact that  $x^2 = x \cdot x = (-x) \cdot (-x) = (-x)^2$  follows from (iii) in Paragraph 2.5).

Finally, we want to prove that  $x^2 > 0$  whenever  $x \neq 0$ . Assume by contradiction that there exists  $x \neq 0$  such that  $x^2 = 0$ . Since  $K$  is a field and  $x \neq 0$ ,  $x^{-1}$  (the inverse of  $x$  for multiplication) exists. Therefore, recalling (i) in Paragraph 2.5 and using that  $x^2 = 0$ , we get

$$0 = 0 \cdot x^{-1} = x^2 \cdot x^{-1} = x \cdot x \cdot x^{-1} = x \cdot 1 = x,$$

hence  $x = 0$ . This contradicts our assumption  $x \neq 0$  and concludes the proof.

- (g) *It holds  $0 < 1$ .*

Proof:  $1 = 1^2 \geq 0$  by inference (f) and  $1 \neq 0$  (see Remark 2.6).

- (h) If  $0 \leq x$  and  $y \leq z$ , then  $xy \leq xz$ .

Proof: Using inference (d), according to which  $z - y \geq 0$ , and the multiplicative compatibility in Definition 2.9,  $xz - xy = x(z - y) \geq 0$  holds, and thus the statement follows in turn from inference (d).

- (i) If  $x \leq 0$  and  $y \leq z$ , then  $xy \geq xz$ .

Proof: Note that  $-x \geq 0$  by inference (e), and  $z - y \geq 0$  by inference (d). Thus, by the multiplicative compatibility in Definition 2.9, we get

$$xy - xz = x(y - z) = (-x)(-(y - z)) = (-x)(z - y) \geq 0.$$

- (j) If  $0 < x \leq y$ , then  $0 < y^{-1} \leq x^{-1}$ .

Proof: We first assert that  $x^{-1} > 0$  ( $y^{-1} > 0$  follows analogously). Indeed, if not, because of  $x^{-1} \neq 0$  then trichotomy in inference (a) implies that  $x^{-1} < 0$ . Accordingly,  $1 = xx^{-1} < 0$  would hold by inference (h), which contradicts (g).

Since  $x^{-1} > 0$  and  $y^{-1} > 0$ , we deduce that  $x^{-1}y^{-1} > 0$ . Therefore, using (h),

$$y^{-1} = xx^{-1}y^{-1} \leq yx^{-1}y^{-1} = x^{-1}.$$

- (k) If  $0 \leq x \leq y$  and  $0 \leq z \leq w$ , then  $0 \leq xz \leq yw$ .

Proof: Exercise 2.12.

- (l) If  $x + y \leq x + z$ , then  $y \leq z$ .

Proof: Exercise 2.12.

- (m) If  $xy \leq xz$  and  $x > 0$ , then  $y \leq z$ .

Proof: Exercise 2.12.

EXERCISE 2.12. — Prove the inferences (k),(l),(m). What happens in (m) when you drop the condition  $x > 0$ , that is, when  $x < 0$  or  $x = 0$ ? For some of the above inferences, formulate and prove similar versions for the strict relation “ $<$ ”.

2.13. — Let  $(K, \leq)$  be an ordered field. As usual, we write  $2, 3, 4, \dots$  for the elements of  $K$  given by  $2 = 1 + 1$ ,  $3 = 2 + 1$ , et cetera. By the compatibility of  $+$  and  $\leq$  in Definition 2.9, and recalling property (g) in Paragraph 2.11, the inequalities

$$\dots < -2 < -1 < 0 < 1 < 2 < 3 < 4 < \dots$$

hold in  $K$ . In particular, the elements  $\dots, -2, -1, 0, 1, 2, 3, \dots$  of  $K$  are all distinct. We identify the set  $\mathbb{Z}$  of integers with a subset of  $K$ . That is, we call the elements  $\{\dots, -2, -1, 0, 1, 2, 3, \dots\}$  of  $K$  “integers”. Consequently, we call elements  $\{pq^{-1} \mid p, q \in \mathbb{Z}, q \neq 0\}$  in  $K$  rational numbers

and thus identify  $\mathbb{Q}$  with a subfield of  $K$ :

$$\mathbb{Z} \subsetneq \mathbb{Q} \subseteq K.$$

In other words, if  $(K, \leq)$  is an ordered field, then it always includes a copy of the rationals inside it.

The above axioms, inferences, and statements in the exercises represent the usual properties for inequalities. We can also use them to solve problems like the one in the following exercise:

EXERCISE 2.14. — Show that

$$\{x \in \mathbb{R} \setminus \{0\} \mid x + \frac{3}{x} + 4 \geq 0\} = \{x \in \mathbb{R} \setminus \{0\} \mid -3 \leq x \leq -1 \text{ or } x > 0\}.$$

*Hint:* note that  $x + \frac{3}{x} + 4 = \frac{(x+3)(x+1)}{x}$ .

#### INTERLUDE: FUNCTIONS

A **function**  $f$  from a set  $X$  to a set  $Y$  is a map that assigns to each  $x \in X$  a uniquely determined element  $y = f(x) \in Y$ . We write  $f : X \rightarrow Y$  for a function from  $X$  to  $Y$  and sometimes also speak of a **mapping** or a **transformation**. We refer to the set  $X$  as **domain**, and the set  $Y$  as **domain of values** or **codomain**.

The set  $F = \{(x, f(x)) \mid x \in X\}$  is called the **graph** of  $f$ . In the context of a function  $f : X \rightarrow Y$ , an element  $x$  of the domain of definition is also called **argument**, and an element  $y = f(x) \in Y$  assumed by the function is also called **value** of the function. If  $f : X \rightarrow Y$  is a function, one also writes

$$\begin{aligned} f : X &\rightarrow Y \\ x &\mapsto f(x), \end{aligned}$$

where  $f(x)$  could be a concrete formula. For example,  $f : \mathbb{R} \rightarrow \mathbb{R}$  with  $x \mapsto x^2$  is a fully defined function in this notation. We pronounce “ $\mapsto$ ” as “is mapped to”. Two functions  $f_1 : X_1 \rightarrow Y_1$  and  $f_2 : X_2 \rightarrow Y_2$  are said to be equal if  $X_1 = X_2$ ,  $Y_1 = Y_2$ , and  $f_1(x) = f_2(x)$  for all  $x \in X_1$ .

DEFINITION 2.15. — Let  $(K, \leq)$  be an ordered field. The **absolute value** or **modulus** on  $K$  is the function  $|\cdot| : K \rightarrow K$  given by

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0. \end{cases}$$

The **sign** is the function  $\text{sgn} : K \rightarrow \{-1, 0, 1\}$  given by

$$\text{sgn}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}.$$

2.16. — In what follows, let  $(K, \leq)$  always be an ordered field, and  $x, y, z, w$  denote elements from  $K$ .

(a) *It holds  $x = \text{sgn}(x) \cdot |x|$ , as well as  $|-x| = |x|$  and  $\text{sgn}(-x) = -\text{sgn}(x)$ .*

(b) *It holds  $|x| \geq 0$ , and  $|x| = 0$  exactly when  $x = 0$ .*

Proof: This follows from the trichotomy property.

(c) (Multiplicativity) *It holds  $\text{sgn}(xy) = \text{sgn}(x)\text{sgn}(y)$  and  $|xy| = |x||y|$ .*

To prove this, check all possible four cases (depending on whether  $x, y$  are negative or not).

(d) *If  $x \neq 0$ , then  $|x^{-1}| = |x|^{-1}$  holds.*

Proof: This follows from (c) because of  $|x^{-1}||x| = 1$ .

(e)  *$|x| \leq y$  is equivalent to  $-y \leq x \leq y$ .*

Proof: First we note that, in both cases,  $y$  is a non-negative number.

Suppose first  $|x| \leq y$ . If  $x \geq 0$  then  $-y \leq 0 \leq x = |x| \leq y$ .

If  $x < 0$  then  $-y \leq -|x| = x < 0 \leq y$  and so again  $-y \leq x \leq y$ .

Suppose now  $-y \leq x \leq y$ . If  $x \geq 0$  then  $|x| = x \leq y$ .

If  $x < 0$  then we observe that the inequality  $-y \leq x$  is equivalent to  $-x \leq y$ . Hence  $|x| = -x \leq y$  also in this case.

(f)  *$|x| < y$  is equivalent to  $-y < x < y$ .*

This is proved by arguing as in (e).

(g) (Triangle inequality) *It holds that*

$$|x + y| \leq |x| + |y|.$$

Proof: Note that, by (e), we have  $-|x| \leq x \leq |x|$  and  $-|y| \leq y \leq |y|$ . Adding these two inequalities, we get

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

So, by property (e),  $|x + y| \leq |x| + |y|$ .

(h) (Inverse triangle inequality) *It holds that  $||x| - |y|| \leq |x - y|$ .*

Proof: The triangle inequality in (g) shows that  $|x| \leq |x - y + y| \leq |x - y| + |y|$ , which



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leads to  $|x| - |y| \leq |x - y|$ . Exchanging  $x$  and  $y$ , we get  $|y| - |x| \leq |x - y|$ . So, by property (e),  $||x| - |y|| \leq |x - y|$ , as desired.

EXERCISE 2.17. — For which  $x, y \in \mathbb{R}$  does equality hold in the triangle inequality? And in the inverse triangle inequality?

### 2.1.2 Axiom of Completeness

To do calculus, ordered fields are generally unsuitable because you can have “gaps”. Indeed, think of the rational numbers  $\mathbb{Q}$  in  $\mathbb{R}$ . The field of rational numbers with the usual order is ordered. So we need more properties, or another axiom, to do calculus in an ordered field. This axiom is the so-called “completeness axiom”. To a certain extent, the search for this axiom began with the work of Greeks such as Pythagoras, Euclid, and Archimedes, but it did not become successful until the 19th century in the work of numerous mathematicians, including Weierstrass, Heine, Cantor, and Dedekind (see also this [link](#)).

#### DEFINITION 2.18: COMPLETENESS AXIOM

Let  $(K, \leq)$  be an ordered field. We say  $(K, \leq)$  is **complete** or a **completely ordered field** if the statement (V) is true.

(V) Let  $X, Y$  be non-empty subsets of  $K$  such that for all  $x \in X$  and  $y \in Y$  the inequality  $x \leq y$  holds. Then there exists  $c \in K$  lying between  $X$  and  $Y$ , in the sense that for all  $x \in X$  and  $y \in Y$  the inequality  $x \leq c \leq y$  holds.

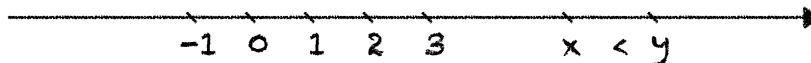
We call statement (V) the **completeness axiom**.

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#### DEFINITION 2.19: REAL NUMBERS

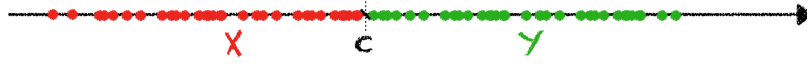
We call **field of real numbers** any completely ordered field. Such a field is denoted with the symbol  $\mathbb{R}$ .

2.20. — We will often visualise the real numbers as the points on a straight line, which is why we also call it the **number line**.



We interpret the relation  $x < y$  for  $x, y \in \mathbb{R}$  as “on the straight line, the point  $y$  lies to the right of the point  $x$ ”. What does the completeness axiom mean in this picture?

Let  $X, Y$  be non-empty subsets of  $\mathbb{R}$  such that for all  $x \in X$  and all  $y \in Y$  the inequality  $x \leq y$  holds. Then all elements of  $X$  are to the left of all elements of  $Y$  as in the following figure.



So, according to the completeness axiom, there exists a number  $c$  that lies in between. The existence of the number  $c$  is, in a sense, an assurance that  $\mathbb{R}$  has no “gaps”. It is advisable to visualize definitions, statements, and their proofs on the number line. However, the number line should always be used only as a motivation and to develop a good intuition, but not for rigorous proof.

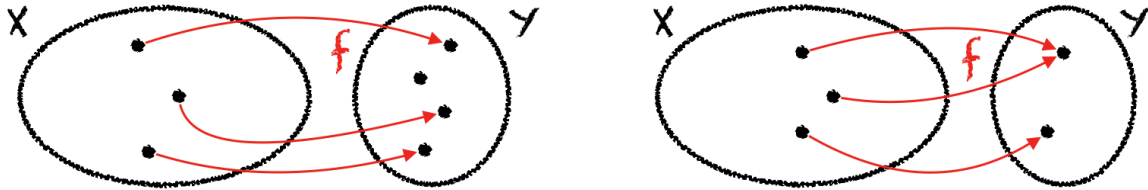
#### INTERLUDE: INJECTIVE, SURJECTIVE AND BIJECTIVE FUNCTIONS

Let  $f : X \rightarrow Y$  be a function. We call  $f$ :

1. **injective** or an **injection** if  $f(x_1) = f(x_2) \implies x_1 = x_2$  for all  $x_1, x_2 \in X$ ,
2. **surjective** or a **surjection** if for every  $y \in Y$  there exists an  $x \in X$  with  $f(x) = y$  and
3. **bijective** or a **bijection** if it is both surjective and injective.

Thus, a function  $f : X \rightarrow Y$  is not injective if there exist two distinct elements  $x_1 \neq x_2 \in X$  with  $f(x_1) = f(x_2)$ , and not surjective if there exists a  $y \in Y$  such that  $f(x) \neq y$  holds for all  $x \in X$ .

In the following image, an injective function that is not surjective is shown on the left, and a surjective function that is not injective is shown on the right.



#### INTERLUDE: IMAGE AND PREIMAGE OF A FUNCTION

DEFINITION 2.21. — For a function  $f : X \rightarrow Y$  and a subset  $A \subset X$  we write

$$f(A) = \{y \in Y \mid \exists x \in A : f(x) = y\}$$

and call this subset of  $Y$  the **image** of  $A$  under the function  $f$ . For a subset  $B \subset Y$  we write

$$f^{-1}(B) = \{x \in X \mid \exists y \in B : f(x) = y\}$$

and call this subset of  $X$  the **preimage** of  $B$  under the function  $f$ .

REMARK 2.22. — Saying that a function  $f : X \rightarrow Y$  is surjective is equivalent to saying that  $f(X) = Y$  (i.e., every element of  $Y$  is in the image of  $X$  under  $f$ ).

EXAMPLE 2.23. — Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be the constant function  $x \mapsto 0$  for every  $x \in \mathbb{R}$ . Then  $f^{-1}(\{0\}) = \mathbb{R}$ , while  $f^{-1}(\{y\}) = \emptyset$  for every  $y \neq 0$ .

EXAMPLE 2.24. — Let  $X, Y$  be two finite sets with the same number of elements (for example,  $X$  and  $Y$  could be the same set). Then, for a function  $f : X \rightarrow Y$ , injectivity and surjectivity are equivalent.

To show this, assume that  $X$  and  $Y$  have  $n$  elements and write  $X = \{x_1, \dots, x_n\}$ . Suppose first that  $f$  is injective. Then all the elements  $f(x_i)$  are distinct, which means that the set  $f(X) = \{f(x_1), \dots, f(x_n)\}$  also has  $n$  elements. Being  $f(X)$  a subset of  $Y$  and  $Y$  has  $n$  elements, the only option is that  $f(X) = Y$ . This proves that injectivity implies surjectivity. Conversely, to show that surjectivity implies injectivity, we prove that if  $f$  is not injective then  $f$  is not surjective. So, assume there exist at least two elements  $x_i \neq x_j$  such that  $f(x_i) = f(x_j)$ . This means that  $f(X)$  has at most  $n - 1$  elements, so  $f$  cannot be surjective.

REMARK 2.25. — For infinite sets, injectivity and surjectivity are not necessarily equivalent. Consider for instance the functions  $f_1, f_2 : \mathbb{N} \rightarrow \mathbb{N}$  defined as

$$f_1(n) = n + 1, \quad f_2(n) = \begin{cases} 0 & \text{if } n = 0, \\ n - 1 & \text{if } n \geq 1. \end{cases}$$

Then  $f_1$  is injective but not surjective, while  $f_2$  is surjective but not injective.

EXERCISE 2.26. — Reformulate the definitions of injectivity, surjectivity, and bijectivity using the notions of image and preimage of a function.

We conclude this section by introducing the **square root function** on  $\mathbb{R}_{\geq 0} = \{x \in \mathbb{R} : x \geq 0\}$  as an application of the completeness axiom. We formulate this as an exercise for the reader.

EXERCISE 2.27. — In this exercise, we show the existence and uniqueness of a bijective function  $\sqrt{\cdot} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$  with property  $(\sqrt{a})^2 = a$  for all  $a \in \mathbb{R}_{\geq 0}$ .

1. Show that, for all  $x, y \in \mathbb{R}_{\geq 0}$ :  $x < y$  is equivalent to  $x^2 < y^2$ .
2. Use Step 1 to deduce that, for every  $a \in \mathbb{R}_{\geq 0}$ , there can exist at most one element  $c \in \mathbb{R}_{\geq 0}$  satisfying  $c^2 = a$ .
3. For a real number  $a \in \mathbb{R}_{\geq 0}$  consider the non-empty subsets

$$X = \{x \in \mathbb{R}_{\geq 0} \mid x^2 \leq a\}, \quad Y = \{y \in \mathbb{R}_{\geq 0} \mid y^2 \geq a\},$$

and apply the completeness axiom to find  $c \in \mathbb{R}$  with  $x \leq c \leq y$  for all  $x \in X$  and  $y \in Y$ . Prove that  $c \in X$  and  $c \in Y$  to conclude that both  $c^2 \leq a$  and  $c^2 \geq a$  hold, thus  $c^2 = a$ .

*Hint:* If by contradiction  $c \notin X$  (that is,  $c^2 > a$ ), then one can find a suitably small real number  $\varepsilon > 0$  such that  $(c - \varepsilon)^2 \geq a$ . Thus  $c - \varepsilon \in Y$ , which contradicts  $y \geq c$  for every  $y \in Y$ . The case of  $c \notin Y$  is analogous.

We call **square root function** the function  $\sqrt{\cdot} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$  that assigns to each  $a \in \mathbb{R}_{\geq 0}$  the number  $c \in \mathbb{R}_{\geq 0}$  uniquely determined by the above construction. We note that  $c^2 = a$ , and we call  $c = \sqrt{a}$  the **square root** of  $a$ . Show that:

4. The function  $\sqrt{\cdot}$  is increasing: for  $x, y \in \mathbb{R}_{\geq 0}$  with  $x < y$ , the inequality  $\sqrt{x} < \sqrt{y}$  holds.
5. The function  $\sqrt{\cdot} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$  is bijective.
6. For all  $x, y \in \mathbb{R}_{\geq 0}$ ,  $\sqrt{xy} = \sqrt{x}\sqrt{y}$ .

EXERCISE 2.28. — For all  $x \in \mathbb{R}$ , show that  $x^2 = |x|^2$  and  $\sqrt{x^2} = |x|$ .

2.29. — In summary, in a field of real numbers as defined in Definition 2.19, the usual arithmetic rules and equation transformations work, although (as usual) division by zero is not defined. Furthermore, the relations  $\leq$  and  $<$  satisfy the usual transformation laws for inequalities. In particular, when multiplying by negative numbers, the inequalities must be reversed. We will use these laws in the following without reference. We will see the deep meaning of the completeness axiom when we use it for further statements. In particular, until further notice, we will always refer to it when we use it.

It is not clear for the moment that there is indeed a field of real numbers as in Definition 2.19. The fact that we occasionally even speak of *the* real numbers stems from the fact that, except for certain identifications, there is only one completely ordered field. In this course, we assume, in agreement with your high school experience, that a field of real numbers exists and is unique.

### 2.1.3 Intervals

DEFINITION 2.30. — Let  $a, b \in \mathbb{R}$ . We define:

- the **closed interval**  $[a, b]$  as

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$

- The **open interval**  $(a, b)$  as

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

- The **half-open intervals**  $[a, b)$  and  $(a, b]$ :

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\} \quad \text{and} \quad (a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}.$$

- The **unbounded closed intervals**

$$[a, \infty) = \{x \in \mathbb{R} \mid a \leq x\} \quad \text{and} \quad (-\infty, b] = \{x \in \mathbb{R} \mid x \leq b\}$$

as well as the **unbounded open intervals**

$$(a, \infty) = \{x \in \mathbb{R} \mid a < x\} \quad \text{and} \quad (-\infty, b) = \{x \in \mathbb{R} \mid x < b\}.$$

2.31. — The intervals  $(a, b]$ ,  $[a, b)$ ,  $(a, b)$  for  $a, b \in \mathbb{R}$  are non-empty exactly when  $a < b$ , and  $[a, b]$  is non-empty exactly when  $a \leq b$ . If the interval is non-empty,  $a$  is called the **left endpoint**,  $b$  is called the **right endpoint** and  $b - a$  is called the **length of the interval**. Intervals of the kind  $[a, b]$ ,  $(a, b]$ ,  $[a, b)$ ,  $(a, b)$  for  $a, b \in \mathbb{R}$  are also called **bounded intervals** if we want to distinguish them from the unbounded intervals.

Instead of round brackets, inverted square brackets are sometimes used to denote open and half-open intervals. For example, instead of  $(a, b)$  for  $a, b \in \mathbb{R}$ , one can also find  $]a, b[$  in the literature. We will always use round brackets here.

#### INTERLUDE: SET OPERATIONS

Let  $P$  and  $Q$  be sets. The **intersection**  $P \cap Q$ , the **union**  $P \cup Q$ , the **relative complement**  $P \setminus Q$ , and the **symmetric difference**  $P \Delta Q$  are defined as by

$$P \cap Q = \{x \mid x \in P \text{ and } x \in Q\}$$

$$P \cup Q = \{x \mid x \in P \text{ or } x \in Q\}$$

$$P \setminus Q = \{x \mid x \in P \text{ and } x \notin Q\}$$

$$P \Delta Q = (P \cup Q) \setminus (P \cap Q).$$

These definitions are illustrated in the following pictures. Sketches of this kind are called **Venn diagrams**.

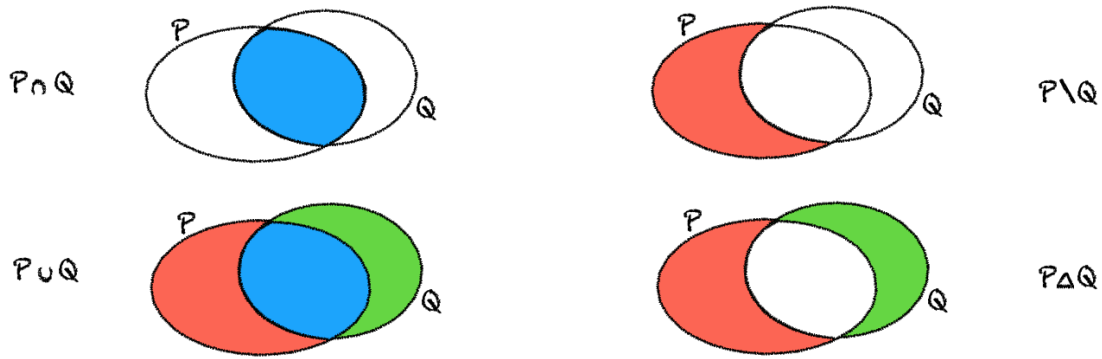
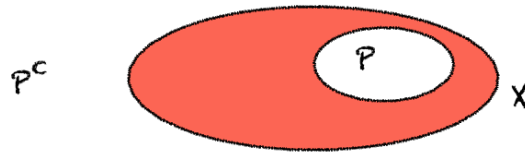


Figure 2.1: Illustration of set operations.

If it is clear from the context that all sets under consideration are subsets of a given basic set  $X$ , then the **complement**  $P^c$  of  $P$  is defined by  $P^c = X \setminus P$ .

Figure 2.2: The complement  $P^c = X \setminus P$  of  $P$  in  $X$ .

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#### INTERLUDE: UNION AND INTERSECTION OF SEVERAL SETS

Let  $\mathcal{A}$  be a family of sets, that is, a set whose elements are themselves sets. Then we define the **union** and the **intersection** of the sets in  $\mathcal{A}$  as

$$\bigcup_{A \in \mathcal{A}} A = \{x \mid \exists A \in \mathcal{A} : x \in A\}, \quad \bigcap_{A \in \mathcal{A}} A = \{x \mid \forall A \in \mathcal{A} : x \in A\}.$$

If  $\mathcal{A} = \{A_1, A_2, \dots\}$ , then we also write

$$\bigcup_{n=1}^{\infty} A_n = \{x \mid \exists n \geq 1 : x \in A_n\}, \quad \bigcap_{n=1}^{\infty} A_n = \{x \mid \forall n \geq 1 : x \in A_n\}$$

for the union and the intersection of the sets in  $\mathcal{A}$ .

- EXERCISE 2.32. — 1. Show that a finite intersection of intervals is again an interval. Can you describe the endpoints of a non-empty intersection using the endpoints of the original intervals?
2. When is a union of two intervals an interval again? In this case, what happens when you unite two intervals of the same type (open, closed, half-open)?

DEFINITION 2.33. — Let  $x \in \mathbb{R}$ . A neighbourhood of  $x$  is a set containing an open interval  $I$  such that  $x \in I$ . Given  $\delta > 0$ , the open interval  $(x - \delta, x + \delta)$  is called the  $\delta$ -neighbourhood of  $x$ .

2.34. — For example, both  $[-1, 1]$  and  $\mathbb{Q} \cup [-1, 1]$  are neighborhoods of  $0 \in \mathbb{R}$  (since they both contain, for instance,  $(-1/2, 1/2)$ ), but  $[0, 1]$  is not a neighbourhood of  $0$ .

We note further that, for  $\delta > 0$  and  $x \in \mathbb{R}$ , the  $\delta$ -neighbourhood of  $x$  is given by  $\{y \in \mathbb{R} \mid |x - y| < \delta\}$ . We will interpret  $|x - y|$  as the **distance** from  $x$  to  $y$ . In terms of “distance”, a few of the above inferences can be re-expressed more intuitively. For example, property (a) in Paragraph 2.16 implies that, for  $x, y \in \mathbb{R}$ , the equality  $|x - y| = |-(x - y)| = |y - x|$  holds. In other words, the distance from  $x$  to  $y$  is equal to the distance from  $y$  to  $x$ .

DEFINITION 2.35: OPEN AND CLOSED SETS

A subset  $U \subseteq \mathbb{R}$  is called **open** in  $\mathbb{R}$  if for every  $x \in U$  there exists an open interval  $I$  such that  $x \in I$  and  $I \subset U$ .

A subset  $F \subseteq \mathbb{R}$  is called **closed** in  $\mathbb{R}$  if its complement  $\mathbb{R} \setminus F$  is open.

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2.36. — Open intervals are open, closed intervals are closed. Intuitively, a subset is open if, for any point  $x$  in the set, all points close enough to  $x$  are also in the set. Contrary to conventional usage, “open” is not the opposite of “closed”.

The sets  $\emptyset$  and  $\mathbb{R}$  are both open in  $\mathbb{R}$ . Hence, they are also closed since  $\emptyset = \mathbb{R}^c$  and  $\mathbb{R} = \emptyset^c$ .

We note that  $\mathbb{Q} \subseteq \mathbb{R}$  and  $[a, b) \subset \mathbb{R}$  are neither open nor closed.

EXERCISE 2.37. — Show that a subset  $U \subseteq \mathbb{R}$  is open exactly if, for every element  $x \in U$ , there exists  $\delta > 0$  such that  $(x - \delta, x + \delta) \subseteq U$ .

EXERCISE 2.38. — Let  $\mathcal{U}$  be a family of open sets, and  $\mathcal{F}$  be a family of closed subsets of  $\mathbb{R}$ . Show that the union and the intersection

$$\bigcup_{U \in \mathcal{U}} U \quad \text{and} \quad \bigcap_{F \in \mathcal{F}} F$$

are open and closed, respectively.

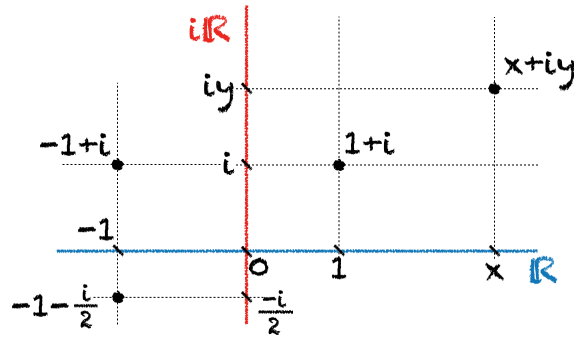
## 2.2 Complex Numbers

### 2.2.1 Definition of Complex Numbers

Using a field of real numbers  $\mathbb{R}$ , we can define the set of complex numbers as

$$\mathbb{C} = \mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We call elements  $z = (x, y) \in \mathbb{C}$  complex numbers, and will write them in the form  $z = x + iy$ , where the symbol  $i$  is called the **imaginary unit**. Note that in this identification, the symbol  $+$  is, for the time being, to be understood as a substitute for the comma. The number  $x \in \mathbb{R}$  is called the **real part** of  $z$  and one writes  $x = \operatorname{Re}(z)$ ; the number  $y \in \mathbb{R}$  is the **imaginary part** of  $z$  and one writes  $y = \operatorname{Im}(z)$ . The elements of  $\mathbb{C}$  with imaginary part 0 are also called **real**, and the elements with real part 0 are called **purely imaginary**. Via the injective mapping  $x \in \mathbb{R} \mapsto x + i0 \in \mathbb{C}$  we identify  $\mathbb{R}$  with the subset of real elements of  $\mathbb{C}$ .



The graphical representation of the set  $\mathbb{C}$  is called the **complex plane** or also **Gaussian number plane**. From this geometric point of view, the set of real points is called the **real axis** and the set of purely imaginary points is called the **imaginary axis**.

As you might expect from previous knowledge,  $i$  should correspond to a square root of  $-1$ . Hence, we want to define an addition and a multiplication on the set  $\mathbb{C}$  so that the set  $\mathbb{C}$  together with these operations is a field in which  $i^2 = -1$  holds. Note that, if  $i^2 = -1$ , then it follows from commutativity and distributivity that

$$(x_1 + iy_1) \cdot (x_2 + iy_2) = x_1x_2 + ix_1y_2 + iy_1x_2 + i^2y_1y_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2).$$

This leads us to the following definition:

**DEFINITION 2.39.** — We call **addition** and **multiplication** on the set  $\mathbb{C} = \mathbb{R} \times \mathbb{R}$  the following operations:

$$\begin{aligned} (x_1, y_1) + (x_2, y_2) &= (x_1 + x_2, y_1 + y_2) && \text{(addition)} \\ (x_1, y_1) \cdot (x_2, y_2) &= (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1) && \text{(multiplication)} \end{aligned}$$



PROPOSITION 2.40. — *The set  $\mathbb{C} = \mathbb{R} \times \mathbb{R}$ , together with the zero element  $(0, 0)$ , the one element  $(1, 0)$ , and the operations introduced in Definition 2.39, is a field.*

*Proof.* We review the axioms of fields. The associativity and commutativity of the addition, and the fact that  $(0, 0)$  is a neutral element for the addition, are direct consequences of the corresponding properties of the addition of real numbers. The inverse element of  $(x, y)$  for the addition is given by  $(-x, -y)$ , since

$$(x, y) + (-x, -y) = (x - x, y - y) = (0, 0).$$

Proving the properties of multiplication requires a little more effort. We start with associativity of multiplication: let  $(x_1, y_1)$ ,  $(x_2, y_2)$ , and  $(x_3, y_3)$  be elements of  $\mathbb{C}$ . Now calculate

$$\begin{aligned} ((x_1, y_1) \cdot (x_2, y_2)) \cdot (x_3, y_3) &= (x_1x_2 - y_1y_2, x_1y_2 + y_1x_2) \cdot (x_3, y_3) \\ &= (x_1x_2x_3 - y_1y_2x_3 - x_1y_2y_3 - y_1x_2y_3, x_1y_2x_3 + y_1x_2x_3 + x_1x_2y_3 - y_1y_2y_3). \end{aligned}$$

Analogously, we calculate

$$\begin{aligned} (x_1, y_1) \cdot ((x_2, y_2) \cdot (x_3, y_3)) &= (x_1, y_1) \cdot (x_2x_3 - y_2y_3, x_2y_3 + y_2x_3) \\ &= (x_1x_2x_3 - y_1y_2x_3 - x_1y_2y_3 - y_1x_2y_3, x_1y_2x_3 + y_1x_2x_3 + x_1x_2y_3 - y_1y_2y_3). \end{aligned}$$

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This proves that the multiplication in Definition 2.39 is associative.

The commutativity of multiplication can also be shown by direct computation:

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1) = (x_2, y_2) \cdot (x_1, y_1).$$

Also, in the same way, we check that  $(1, 0)$  is the neutral element for multiplication:

$$(x, y) \cdot (1, 0) = (x \cdot 1 - y \cdot 0, x \cdot 0 + y \cdot 1) = (x, y).$$

Next, we check the distributivity law: Let again  $(x_1, y_1)$ ,  $(x_2, y_2)$ , and  $(x_3, y_3)$  be elements of  $\mathbb{C}$ . Then

$$\begin{aligned} (x_1, y_1) \cdot ((x_2, y_2) + (x_3, y_3)) &= (x_1, y_1) \cdot (x_2 + x_3, y_2 + y_3) \\ &= (x_1x_2 + x_1x_3 - y_1y_2 - y_1y_3, y_1x_2 + y_1x_3 + x_1y_2 + x_1y_3) \\ &= (x_1x_2 - y_1y_2, y_1x_2 + x_1y_2) + (x_1x_3 - y_1y_3, y_1x_3 + x_1y_3) \\ &= (x_1, y_1) \cdot (x_2, y_2) + (x_1, y_1) \cdot (x_3, y_3), \end{aligned}$$

which shows that  $\mathbb{C}$  is a ring when endowed with the addition and multiplication given in Definition 2.39.

To finish the proof, we still need to show the existence of multiplicative inverses. Let  $(x, y) \in \mathbb{C}$  be such that  $(x, y) \neq (0, 0)$ . So either  $x \neq 0$  or  $y \neq 0$  holds, and therefore

$x^2 + y^2 > 0$ . Then the multiplicative inverse of  $(x, y)$  is given by  $\left(\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2}\right)$ , because

$$\begin{aligned} (x, y) \cdot \left(\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2}\right) &= \left(x \cdot \frac{x}{x^2+y^2} - y \cdot \frac{-y}{x^2+y^2}, y \cdot \frac{x}{x^2+y^2} + x \cdot \frac{-y}{x^2+y^2}\right) \\ &= \left(\frac{x^2+y^2}{x^2+y^2}, \frac{yx-xy}{x^2+y^2}\right) = (1, 0). \end{aligned}$$

□

**Applet 2.41 (Complex numbers).** We consider the field operations (addition, multiplication, multiplicative inverse) on the complex numbers. The true geometric meaning of multiplication and of the multiplicative inverse will be discussed later.

As already explained, we do not write  $(x, y)$  for complex numbers, but  $x + iy$ . Instead of  $x + i0$  we also simply write  $x$ , and instead of  $0 + iy$  we write  $iy$ , and finally we also write  $i$  for  $i1$ . By construction,  $i^2 = -1$  holds. According to this notation, take  $\mathbb{R}$  to be a subset of  $\mathbb{C}$ . This makes sense since addition and multiplication on  $\mathbb{C}$  restricted to  $\mathbb{R}$  is equivalent to addition and multiplication on  $\mathbb{R}$ . Also, given  $z, w \in \mathbb{C}$ , we write  $zw$  for  $z \cdot w$ .

Given  $z \neq 0$ , we shall use both  $z^{-1}$  and  $\frac{1}{z}$  to denote the inverse of  $z$  for the multiplication. For instance,  $i^{-1} = \frac{1}{i} = -i$  (since  $(-i) \cdot i = -i^2 = 1$ ).

**DEFINITION 2.42.** — Let  $z = x + iy$  be a complex number. We call  $\bar{z} = x - iy$  the complex number **conjugate** to  $z$ . The mapping  $\mathbb{C} \rightarrow \mathbb{C}$  given by  $z \mapsto \bar{z}$  is called **complex conjugation**.

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**LEMMA 2.43.** — The complex conjugation satisfies the following properties:

1. For all  $z \in \mathbb{C}$ ,  $z\bar{z} \in \mathbb{R}$  and  $z\bar{z} \geq 0$ . Furthermore, for all  $z \in \mathbb{C}$ ,  $z\bar{z} = 0$  holds exactly when  $z = 0$ .
2. For all  $z, w \in \mathbb{C}$ ,  $\overline{z+w} = \bar{z} + \bar{w}$  holds.
3. For all  $z, w \in \mathbb{C}$ ,  $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$  holds.

*Proof.* Part (1) follows from the fact that, for  $z = x + iy$ ,

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2.$$

To show parts (2) and (3), we write  $z = x_1 + iy_1$  and  $w = x_2 + iy_2$ . Then  $z + w = (x_1 + x_2) + i(y_1 + y_2)$  and we get

$$\overline{z+w} = (x_1 + x_2) - i(y_1 + y_2) = (x_1 - iy_1) + (x_2 - iy_2) = \bar{z} + \bar{w}.$$

Analogously, since  $z \cdot w = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$  we have

$$\overline{z \cdot w} = (x_1x_2 - y_1y_2) - i(x_1y_2 + y_1x_2) = (x_1 - iy_1) \cdot (x_2 - iy_2) = \bar{z} \cdot \bar{w},$$

as desired. □

EXERCISE 2.44. — Show the identities

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2} \quad \text{and} \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$$

for all  $z \in \mathbb{C}$ . In particular, conclude that  $\mathbb{R} = \{z \in \mathbb{C} \mid z = \bar{z}\}$ . Can you interpret these equalities geometrically?

2.45. — Since  $i^2 = -1 < 0$ , property (f) in Paragraph 2.11 implies that no order compatible with addition and multiplication can be defined on  $\mathbb{C}$ . Nevertheless, calculus can be performed on the complex numbers, which is partly addressed in this course but mainly in the course on *complex analysis* in the second year of the study of mathematics and physics. The reason for this is that  $\mathbb{C}$  satisfies a generalisation of the completeness axiom, which we can only discuss after some more theory.

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### 2.2.2 The Absolute Value on the Complex Numbers

Since there is no ordering relation on the field of complex numbers that would make it an ordered field, we cannot use Definition 2.15 to define an absolute value on  $\mathbb{C}$ . We still want to define an absolute value on  $\mathbb{C}$  in such a way that it has as many properties as possible of the absolute value on  $\mathbb{R}$  and is compatible with it. To do this, we use the root function introduced in Exercise 2.27.

DEFINITION 2.46. — The **absolute value** or **norm** on  $\mathbb{C}$  is the function  $|\cdot| : \mathbb{C} \rightarrow \mathbb{R}$  given by

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}$$

for  $z = x + iy \in \mathbb{C}$ .

At this point we note that, given  $x \in \mathbb{R}$ , the absolute value  $|x| = \operatorname{sgn}(x) \cdot x$  and the absolute value of  $x$  as an element of  $\mathbb{C}$  coincide, since  $\sqrt{x\bar{x}} = \sqrt{x^2} = |x|$  holds. In particular, the newly introduced notation is consistent, and we have extended the absolute value of  $\mathbb{R}$  to  $\mathbb{C}$ .

Note that  $|z| \geq 0$  for all  $z \in \mathbb{C}$ , and  $|z| = 0$  exactly when  $z = 0$ . Also, the absolute value on  $\mathbb{C}$  is multiplicative, namely

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$$|zw| = \sqrt{zw\overline{zw}} = \sqrt{z\bar{z}w\bar{w}} = \sqrt{z\bar{z}}\sqrt{w\bar{w}} = |z||w| \quad \text{for all } z, w \in \mathbb{C}.$$

Furthermore,

$$z^{-1} = \frac{\bar{z}}{|z|^2} \quad \text{for all } z \neq 0.$$

These are essential consequences of Lemma 2.43. Finally, the triangle inequality holds, as shown in the next proposition.

PROPOSITION 2.47 (Triangle inequality). — *For all  $z, w \in \mathbb{C}$ ,  $|z + w| \leq |z| + |w|$  holds.*

To prove this result, we first need the following:

LEMMA 2.48 (Cauchy-Schwarz inequality). — *For all  $z = x_1 + iy_1, w = x_2 + iy_2 \in \mathbb{C}$ ,*

$$x_1x_2 + y_1y_2 \leq |z||w|. \quad (2.3)$$

*Proof.* We begin by observing that

$$\begin{aligned} |z|^2|w|^2 - (x_1x_2 + y_1y_2)^2 &= (x_1^2 + y_1^2)(x_2^2 + y_2^2) - (x_1x_2 + y_1y_2)^2 \\ &= x_1^2x_2^2 + y_1^2y_2^2 + y_1^2x_2^2 + x_1^2y_2^2 - (x_1^2x_2^2 + y_1^2y_2^2 + 2x_1x_2y_1y_2) \\ &= y_1^2x_2^2 + x_1^2y_2^2 - 2x_1x_2y_1y_2 \\ &= (y_1x_2 - x_1y_2)^2 \geq 0. \end{aligned}$$

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This proves that

$$(x_1x_2 + y_1y_2)^2 \leq |z|^2|w|^2.$$

Taking the square root on both sides and recalling Exercise 2.28, we get  $|x_1x_2 + y_1y_2| \leq |z||w|$  which implies (2.3) (recall that  $x \leq |x|$  for any  $x \in \mathbb{R}$ ).  $\square$

*Proof of Proposition 2.47.* We write  $z = x_1 + iy_1$  and  $w = x_2 + iy_2$ . Then, since  $z + w = (x_1 + x_2) + i(y_1 + y_2)$ , we have

$$\begin{aligned} |z + w|^2 &= (x_1 + x_2)^2 + (y_1 + y_2)^2 \\ &= x_1^2 + x_2^2 + y_1^2 + y_2^2 + 2(x_1x_2 + y_1y_2) \\ &= |z|^2 + |w|^2 + 2(x_1x_2 + y_1y_2). \end{aligned}$$

Hence, using (2.3), we get

$$|z + w|^2 = |z|^2 + |w|^2 + 2(x_1x_2 + y_1y_2) \leq |z|^2 + |w|^2 + 2|z||w| = (|z| + |w|)^2.$$

Taking the square root on both sides concludes the proof.  $\square$

2.49. — The absolute value of the complex number  $z = x + iy$  is the square root of  $x^2 + y^2$  and, in the geometric notion of complex numbers, is equivalent to the length of the straight

line from the origin  $0 + i0$  to  $z$ . In the same way, for two complex numbers  $z$  and  $w$ , we interpret  $|z - w|$  as the **distance** from  $z$  to  $w$ .

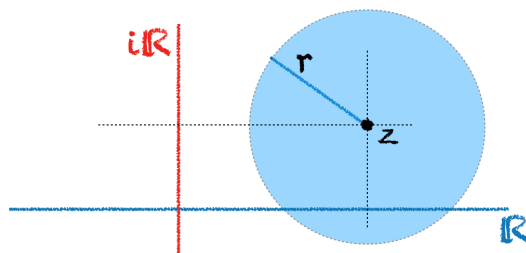
DEFINITION 2.50. — The **open circular disk** with radius  $r > 0$  around a point  $z \in \mathbb{C}$  is the set

$$B(z, r) = \{w \in \mathbb{C} \mid |z - w| < r\}.$$

The **closed circular disk** with radius  $r > 0$  around  $z \in \mathbb{C}$  is the set

$$\overline{B(z, r)} = \{w \in \mathbb{C} \mid |z - w| \leq r\}.$$

2.51. — The open circular disk  $B(z, r)$  thus consists precisely of those points that have distance strictly less than  $r$  from  $z$ . Open circular disks in  $\mathbb{C}$  and open intervals in  $\mathbb{R}$  are compatible in the following sense: If  $x \in \mathbb{R}$  and  $r > 0$ , then the intersection of the open circular disk  $B(x, r) \subseteq \mathbb{C}$  with  $\mathbb{R}$  is just the open interval  $(x - r, x + r)$  lying symmetrically about  $x$ .



EXERCISE 2.52. — Show the following property of open circular disks: let  $z_1, z_2 \in \mathbb{C}$ ,  $r_1 > 0$  and  $r_2 > 0$ . For each point  $z \in B(z_1, r_1) \cap B(z_2, r_2)$  there exists a radius  $r > 0$  such that

$$B(z, r) \subseteq B(z_1, r_1) \cap B(z_2, r_2).$$

Illustrate your choice of radius  $r$  in a picture.

The definition of open set in  $\mathbb{C}$  given below generalizes that in  $\mathbb{R}$  from Exercise 2.37.

DEFINITION 2.53. — A subset  $U \subseteq \mathbb{C}$  is called **open** in  $\mathbb{C}$  if for every point in  $U$  there exists an open circular disk around that point contained in  $U$ . More formally, for all  $z \in U$  there exists a radius  $r > 0$  such that  $B(z, r) \subseteq U$ .

A subset  $A \subseteq \mathbb{C}$  is called **closed** in  $\mathbb{C}$  if its complement  $\mathbb{C} \setminus A$  is open.

For example, thanks to Exercise 2.52, all open circular disks are open. In addition to open circular disks, there are many other subsets of  $\mathbb{C}$ . For example, every union of open subsets is open. We will return to studying open sets and related notions in much greater generality in the second semester.

## 2.3 Maximum and Supremum

### 2.3.1 Existence of the Supremum

#### DEFINITION 2.54: BOUNDED SETS, MAXIMA AND MINIMA

Let  $X \subseteq \mathbb{R}$  be a subset of the real numbers.

- $X$  is **bounded from above** if there exists  $s \in \mathbb{R}$  such that  $x \leq s$  for all  $x \in X$ . Any such  $s \in \mathbb{R}$  is called an **upper bound** of  $X$ . An upper bound that is itself contained in  $X$  is called a **maximum**. We write

$$s = \max(X)$$

if such a maximum of  $X$  exists and is equal to  $s$ .

- The terms **bounded from below**, **lower bound**, and **minimum** are defined analogously.
- $X \subset \mathbb{R}$  is called **bounded** if it is bounded from above and from below.

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2.55. — If a subset  $X \subseteq \mathbb{R}$  has a maximum, then it is unique. Indeed, if  $x_1 \in X$  and  $x_2 \in X$  both satisfy the properties of a maximum, it follows that  $x_1 \leq x_2$  because  $x_2$  is a maximum, and  $x_2 \leq x_1$  because  $x_1$  is a maximum, and hence  $x_1 = x_2$ . So we may speak of *the* maximum of a subset  $X \subset \mathbb{R}$  if one exists.

A closed interval  $[a, b]$  with endpoints  $a < b$  in  $\mathbb{R}$  has  $a = \min([a, b])$  as minimum and  $b = \max([a, b])$  as maximum. However, there are also sets that do not have a maximum. For example, the open interval  $(a, b)$  with endpoints  $a < b$  in  $\mathbb{R}$  has no maximum. Although the endpoint  $b$  would lend itself as a maximum, it does not lie in the set  $(a, b)$  and is therefore not a candidate for the maximum. Likewise  $\mathbb{R}$  as a subset of  $\mathbb{R}$ , or also unbounded intervals of the form  $[a, \infty)$ ,  $(a, \infty)$  for  $a \in \mathbb{R}$ , do not have a maximum.

#### DEFINITION 2.56: SUPREMUM

Let  $X \subseteq \mathbb{R}$  be a subset and let  $A := \{a \in \mathbb{R} \mid x \leq a \text{ for all } x \in X\}$  be the set of all upper bounds of  $X$ . If the minimum  $s = \min(A)$  exists, then we call this minimum the **supremum** of  $X$ , and write

$$s = \sup(X).$$

In other words, if it exists, the supremum of  $X$  is the smallest upper bound of  $X$ . We can describe the supremum  $s = \sup(X)$  of  $X$  directly by

$$x \leq s \text{ for all } x \in X, \quad \text{and} \quad x \leq t \text{ for all } x \in X \implies s \leq t. \quad (2.4)$$

Equivalently, the supremum of  $X$  could also be characterised by the fact that no real number  $x_1$  strictly smaller than  $s = \sup(X)$  is an upper bound of  $X$ :

$$x \leq s \text{ for all } x \in X, \quad \text{and} \quad t < s \implies \exists x \in X \text{ with } t < x. \quad (2.5)$$

The statements (2.4) and (2.5) are equivalent.

We have not yet addressed the question of the existence of the supremum. If  $X$  is the empty set or if  $X$  is unbounded from above, then the supremum cannot exist. In all other cases it exists, as the following theorem shows.

**THEOREM 2.57: EXISTENCE OF SUPREMUM**

*Let  $X \subset \mathbb{R}$  be a nonempty subset bounded from above. Then the supremum of  $X$  exists.*

*Proof.* By assumption,  $X$  is non-empty, and the set of upper bounds  $A := \{a \in \mathbb{R} \mid x \leq a \text{ for all } x \in X\}$  is also non-empty. Furthermore, for all  $x \in X$  and  $a \in A$ , the inequality  $x \leq a$  holds. Therefore, according to the completeness axiom in Definition 2.18, it follows that there exists a  $c \in \mathbb{R}$  such that

$$x \leq c \leq a$$

holds for all  $x \in X$  and  $a \in A$ . From the first inequality it follows that  $c$  is an upper bound of  $X$ , so  $c \in A$ . From the second inequality it follows that  $c$  is the minimum of the set  $A$ .  $\square$

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**Applet 2.58** ([Supremum of a bounded non-empty set](#)). We consider a bounded non-empty subset of  $\mathbb{R}$  and two equivalent characterizations of the supremum of this set.

**PROPOSITION 2.59.** — *Let  $X$  and  $Y$  be nonempty subsets of  $\mathbb{R}$  bounded from above, and write*

$$X + Y := \{x + y \mid x \in X, y \in Y\} \quad \text{and} \quad XY := \{xy \mid x \in X, y \in Y\}.$$

*The sets  $X \cup Y$ ,  $X \cap Y$ , and  $X + Y$  are bounded from above. Also, if  $x \geq 0$  and  $y \geq 0$  for all  $x \in X$  and  $y \in Y$ , then  $XY$  is bounded from above. Furthermore:*

- (1)  $\sup(X \cup Y) = \max\{\sup(X), \sup(Y)\}$ ,
- (2) if  $X \cap Y$  is not empty, then  $\sup(X \cap Y) \leq \min\{\sup(X), \sup(Y)\}$ ,
- (3)  $\sup(X + Y) = \sup(X) + \sup(Y)$  and
- (4) if  $x \geq 0$  and  $y \geq 0$  for all  $x \in X$  and  $y \in Y$ , then  $\sup(XY) = \sup(X) \sup(Y)$ .

*Proof.* The proof of (1) and (2) is left to the reader.

To prove (3) we write  $x_0 = \sup(X)$  and  $y_0 = \sup(Y)$ . Let  $z \in X + Y$ . There are  $x \in X$  and  $y \in Y$  with  $z = x + y$ , and because of  $x \leq x_0$  and  $y \leq y_0$ , then  $z \leq x_0 + y_0$  holds. That

is,  $x_0 + y_0$  is an upper bound for  $X + Y$ . We want to show that  $x_0 + y_0$  is the supremum of  $X + Y$ , i.e., the smallest upper bound of  $X + Y$ .

Let  $z_0 = \sup(X + Y)$  (the supremum exists, since we just proved that  $X + Y$  is bounded from above) and assume by contradiction that

$$\varepsilon = x_0 + y_0 - z_0 > 0.$$

Since  $x_0$  is the supremum of  $X$ , it follows from (2.5) that there exists  $x \in X$  with  $x > x_0 - \varepsilon/2$ , and similarly there exists  $y \in Y$  with  $y > y_0 - \varepsilon/2$ . Set  $z = x + y$ . It follows that

$$z = x + y > x_0 - \varepsilon/2 + y_0 - \varepsilon/2 = z_0,$$

which contradicts the fact that  $z_0$  is an upper bound for  $X + Y$ . Thus  $z_0 = x_0 + y_0$ , as desired.

The proof of (4) is done in a similar way.  $\square$

2.60. — For a non-empty subset  $X \subseteq \mathbb{R}$  bounded from below, the largest lower bound of  $X$  will also be called the **infimum**  $\inf(X)$  of  $X$ . An existence statement analogous to Theorem 2.57 holds for the infimum. Alternatively, the infimum of  $X$  can be written as

$$-\sup\{-x \mid x \in X\}.$$

In this way, practically all statements about infima can be traced back to statements about suprema.

## 2.3.2 Two-point Compactification

2.61. — In this section we want to extend the notions of **supremum** and **infimum** to arbitrary subsets of  $\mathbb{R}$ . To do this, we use the symbols  $\infty = +\infty$  and  $-\infty$ , which are not real numbers. We define the **extended number line**, which is also called **two-point compactification** of  $\mathbb{R}$ , by

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$$

and imagine this as the number line

$$\begin{array}{ccc} \bullet & \text{-----} & \bullet \\ -\infty & & \mathbb{R} & & +\infty \end{array}$$

Here we have added the point  $+\infty$  to the right of  $\mathbb{R}$  and the point  $-\infty$  to the left of  $\mathbb{R}$  to the straight line. We extend the order relation of the real numbers  $\leq$  to  $\overline{\mathbb{R}}$  by requiring  $-\infty < x < +\infty$  for all  $x \in \mathbb{R}$ .



To simplify the notation, we also write  $\infty$  in place of  $+\infty$ . Often used calculation rules for the symbols  $-\infty$  and  $\infty$ , such as the following, are standard conventions:

$$\infty + x = \infty + \infty = \infty \quad \text{and} \quad -\infty + x = -\infty - \infty = -\infty$$

for all  $x \in \mathbb{R}$ . Also, for  $x > 0$ ,

$$x \cdot \infty = \infty \cdot \infty = \infty \quad \text{and} \quad x \cdot (-\infty) = \infty \cdot (-\infty) = -\infty.$$

One should use such conventions as sparingly as possible and be careful with them. The expressions  $\infty - \infty$  and  $0 \cdot \infty$  or similar remain undefined.

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DEFINITION 2.62. — Let  $X$  be a subset of  $\mathbb{R}$ . If  $X \subset \mathbb{R}$  is not bounded from above, then we define  $\sup(X) = \infty$ . If  $X$  is empty, we set  $\sup(\emptyset) = -\infty$ . In this context, we call  $\infty$  and  $-\infty$  **undefinite values**.

In other words, the statement  $\sup(X) = \infty$  is equivalent, by definition, to the statement that  $X$  is not bounded from above, that is,

$$\forall x_0 \in \mathbb{R} \quad \exists x \in X : x > x_0.$$

In the same way  $\sup(X) = -\infty$  is equivalent, by definition, to the statement  $X = \emptyset$ .

Similarly, we define  $\inf(\emptyset) = +\infty$ , and  $\inf(X) = -\infty$  if  $X \subseteq \mathbb{R}$  is not bounded from below.

## 2.4 Consequences of Completeness

We introduced the root function in Section 2.1 using the completeness axiom, and used the completeness axiom in Section 2.3 to prove the existence of the supremum. In this section we will discuss some further consequences of the completeness axiom. We choose for the whole Section 2.4 an arbitrary set of real numbers  $\mathbb{R}$ , and call the elements of  $\mathbb{R}$  real numbers.

### 2.4.1 The Archimedean Principle

Archimedes' principle is the statement that for every real number  $x \in \mathbb{R}$  there is an integer  $n$  greater than  $x$ . The following theorem, which we prove using the existence of the supremum, is a slightly more precise formulation of Archimedes' Principle.

**THEOREM 2.63.** — *For every  $x \in \mathbb{R}$  there exists exactly one  $n \in \mathbb{Z}$  with  $n \leq x < n + 1$ .*

*Proof.* First, let  $x \geq 0$  be a non-negative real number. Then  $E = \{n \in \mathbb{Z} \mid n \leq x\}$  is an upper bounded subset of  $\mathbb{R}$  which is non-empty since  $0 \in E$ . By Theorem 2.57, the supremum  $s_0 = \sup(E)$  exists. Since  $s_0$  is the smallest upper bound of  $E$ , then  $s_0 \leq x$  holds and there exists  $n_0 \in E$  with  $s_0 - 1 < n_0$ . It follows that  $s_0 < n_0 + 1$  and therefore  $m \leq s_0 < n_0 + 1$  for every  $m \in E$ . Thus,  $m \leq n_0$  for every  $m \in E$ , which means that  $n_0$  is the maximum of  $E$  and  $n_0 + 1 \notin E$  (i.e.,  $n_0 + 1 > x$ ). In summary, it follows that

$$n_0 \leq s_0 \leq x < n_0 + 1,$$

which proves Theorem 2.63 in the case  $x \geq 0$ .

If  $x < 0$ , then we can apply the above case to  $-x$  and conclude the existence of  $m \in \mathbb{Z}$  such that  $m \leq -x < m + 1$ , or equivalently  $-m - 1 < x \leq -m$ . If  $x = -m$ , then the result follows with  $n = -m$ . If  $x < -m$ , then the result follows with  $n = -m - 1$ .

For the proof of uniqueness, we assume that for  $n_1, n_2 \in \mathbb{Z}$  both inequalities  $n_1 \leq x < n_1 + 1$  and  $n_2 \leq x < n_2 + 1$  hold. It follows that  $n_1 \leq x < n_2 + 1$  and thus  $n_1 \leq n_2$ . Similarly  $n_2 \leq n_1$ , implying  $n_1 = n_2$ .  $\square$

**DEFINITION 2.64.** — The **integer part**  $\lfloor x \rfloor$  of a number  $x \in \mathbb{R}$  is the integer  $n \in \mathbb{Z}$  uniquely determined by Theorem 2.63 with  $n \leq x < n + 1$ . The function from  $\mathbb{R}$  to  $\mathbb{Z}$  given by  $x \mapsto \lfloor x \rfloor$  is called **rounding function**. The **fractional part** of a real number  $x$  is  $\{x\} = x - \lfloor x \rfloor \in [0, 1)$ .

**COROLLARY 2.65.** — *For every  $\varepsilon > 0$  there exists an integer  $n \geq 1$  such that  $\frac{1}{n} < \varepsilon$  holds.*

*Proof.* If  $\varepsilon > 1$  then the result holds with  $n = 1$ .

If  $\varepsilon \leq 1$  then  $\frac{1}{\varepsilon} \geq 1$  and, thanks to Theorem, 2.63 there exists  $n \geq 1$  with  $n > \frac{1}{\varepsilon}$ . Then, for such  $n$ , the inequality  $\frac{1}{n} < \varepsilon$  holds.  $\square$

**COROLLARY 2.66.** — *For every two real numbers  $a, b \in \mathbb{R}$  with  $a < b$ , there is one  $r \in \mathbb{Q}$  with  $a < r < b$ .*

*Proof.* Set  $\varepsilon = b - a$ . According to Archimedes' principle in the form of Corollary 2.65, there exists  $m \in \mathbb{N}$  with  $\frac{1}{m} < \varepsilon$ .

Similarly, according to Archimedes' principle in the form of Theorem 2.63, there exists  $n \in \mathbb{Z}$  with  $n \leq ma < n + 1$ , or equivalently  $\frac{n}{m} \leq a < \frac{n+1}{m}$ .

Hence, since  $\frac{1}{m} < \varepsilon$  and  $a + \varepsilon = b$ , we get

$$a < \frac{n+1}{m} = \frac{n}{m} + \frac{1}{m} \leq a + \frac{1}{m} < a + \varepsilon = b.$$

This proves the corollary with  $r = \frac{n+1}{m}$ .  $\square$

Stated differently, the above corollary shows that  $\mathbb{Q}$  intersects any open non-empty interval  $I$ , that is,  $I \cap \mathbb{Q} \neq \emptyset$ . A subset  $X$  of  $\mathbb{R}$  is called **dense** in  $\mathbb{R}$  if every open non-empty interval of  $\mathbb{R}$  contains an element of  $X$ . Corollary 2.66 thus states:  $\mathbb{Q}$  is dense in  $\mathbb{R}$ .

The following two exercises establish a generalisation of Archimedes' principle which will be used in the section about decimal fractions.

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**EXERCISE 2.67.** — Let  $A := \{m_0, m_1, m_2, \dots\} \subset \mathbb{Z}$  be a strictly increasing sequence of integers, i.e.,  $m_{i+1} > m_i$  for every  $i \geq 0$ . Then Theorem 2.63 holds with  $A$  in place of  $\mathbb{Z}$ . In other words, for every  $x \in \mathbb{R}$  there exists exactly one element  $m_i \in A$  such that  $m_i \leq x < m_{i+1}$ .

**EXERCISE 2.68.** — Let  $A := \{m_0, m_1, m_2, \dots\} \subset \mathbb{Z}$  be a strictly increasing sequence of positive integers. Then Corollary 2.65 holds with  $A$  in place of  $\mathbb{N}$ . In other words, for every  $\varepsilon > 0$  there exists an element  $m_i \in A$  such that  $\frac{1}{m_i} < \varepsilon$  holds.

## 2.4.2 Decimal Fraction Expansion and Uncountability (Extra material)

2.69. — A common notion of the real numbers is given by non-terminating decimal fractions. Formally, we define a **decimal fraction** as a sequence of integers

$$a_0, a_1, a_2, a_3, \dots$$

with  $a_0 \in \mathbb{Z}$ , and  $0 \leq a_n \leq 9$  for all  $n \geq 1$ . Thus, given a decimal fraction, we can assign to it a unique element of  $\mathbb{R}$  as follows.

Suppose  $a_0 \geq 0$ . We set

$$x_n = \sum_{k=0}^n a_k \cdot 10^{-k} \quad \text{and} \quad y_n = 10^{-n} + \sum_{k=0}^n a_k \cdot 10^{-k}. \quad (2.6)$$

One can check that

$$x_0 \leq x_1 \leq \dots \leq x_n \leq x_{n+1} \leq \dots \leq y_{n+1} \leq y_n \leq \dots \leq y_1 \leq y_0.$$

Thus, if we consider the sets  $X = \{x_0, x_1, x_2, \dots\}$  and  $Y = \{y_0, y_1, y_2, \dots\}$ , according to the completeness axiom we can conclude the existence of  $c \in \mathbb{R}$  with the property that

$$x_n \leq c \leq y_n \quad (2.7)$$

for all  $n \in \mathbb{N}$ . Archimedes' principle in the form of Corollary 2.65 shows that there is precisely one real number  $c$  that satisfies the inequality  $x_n \leq c \leq y_n$  for all  $n \geq 0$ . If there were two different such numbers, say  $c$  and  $d$  with  $c < d$ , it follows from

$$x_n \leq c < d \leq y_n$$

and the definition of  $x_n$  and  $y_n$  the inequality

$$0 < d - c < 10^{-n}$$

for all  $n \geq 0$ , which contradicts Exercise 2.68 with  $m_i = 10^i$ . We call the element  $c \in \mathbb{R}$  uniquely determined by (2.7) the real number with decimal expansion  $a_0, a_1 a_2 a_3 a_4 \dots$ . We note that two possible alternative definitions would be

$$c = \sup\{x_n \mid n \geq 0\} \quad \text{or} \quad c = \inf\{y_n \mid n \geq 0\}.$$

If  $a_0$  is negative, we first consider the real number  $c$  with decimal expansion  $-a_0, a_1 a_2 a_3 a_4 \dots$  and then define the real number with decimal expansion  $a_0, a_1 a_2 a_3 a_4 \dots$  as  $-c$ .

2.70. — Now, the following question arises: Can every element of  $\mathbb{R}$  be written as a decimal fraction? This is indeed the case: Let  $c \in \mathbb{R}$  and  $c \geq 0$ . Then we can write  $a_0 := \lfloor c \rfloor$  and define

$$a_n := \lfloor 10^n c \rfloor - 10 \lfloor 10^{n-1} c \rfloor. \quad (2.8)$$

One verifies that the sequence  $a_0, a_1, a_2, \dots$  is indeed a decimal fraction, since  $0 \leq a_n \leq 9$  is satisfied for all  $n \geq 1$ . Also, it is not difficult to check the inequalities (2.7), from which it follows that  $c$  is just the real number with decimal expansion  $a_0, a_1 a_2 a_3 \dots$ . The case  $c \leq 0$  is obtained in the same way, by changing the sign of  $a_0$  in the decimal expansion of  $-c$ .

2.71. — It is important to remark that two different decimal fractions can represent the same real number. For example, the two decimal fractions

$$0.199999999999\ldots \quad \text{and} \quad 0.20000000000000\ldots$$

both represent the real number  $\frac{1}{5}$ . However, the problem only occurs when a decimal fraction becomes a constant sequence  $\ldots 9999\ldots$  after a certain point. To rule this out, we can consider the following definition: We call **real decimal fraction** any sequence of integers

$$a_0, a_1, a_2, a_3, \ldots$$

with  $0 \leq a_n \leq 9$  for all  $n \geq 1$ , and with the property that for every  $n_0 \geq 1$  there exists a  $n \geq n_0$  with  $a_n \neq 9$ .

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EXERCISE 2.72. — Let  $c \geq 0$  be a real number. Verify that the sequence  $a_0, a_1, a_2, \ldots$  defined by (2.8) is a real decimal fraction. Then show that this gives rise to a bijection between  $\mathbb{R}$  and the set of all real decimal fractions.

#### INTERLUDE: POWER SET

Let  $X$  be a set. The **power set**  $\mathcal{P}(X)$  of  $X$  is the set of all subsets of  $X$ , that is,

$$\mathcal{P}(X) = \{Q \mid Q \text{ is a set and } Q \subset X\}.$$

EXAMPLE 2.73. — If  $X = \{0, 1, 2\}$  then

$$\mathcal{P}(X) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$$

## INTERLUDE: CARDINALITY

Let  $X$  and  $Y$  be two sets. We say that  $X$  and  $Y$  have the **same cardinality** or the **same number of elements**, written  $X \sim Y$ , if there is a bijection  $f : X \rightarrow Y$ . We say that  $Y$  is **larger** than  $X$ , and write  $X \lesssim Y$ , if there is an injection  $f : X \rightarrow Y$ .

- We say that the cardinality of the empty set is zero, and write  $|\emptyset| = 0$ .
- Let  $X$  be a set and  $n \geq 1$  a natural number. We say the set  $X$  has cardinality  $n$ , and write  $|X| = n$ , if there is a bijection from  $X$  to  $\{1, \dots, n\}$ . In this case we call  $X$  a **finite set** and write  $|X| < \infty$ .
- If  $X$  is not finite, we call  $X$  an **infinite set**.
- A set is called **countable** if there is a bijection to  $\mathbb{N}$ . The cardinality of  $\mathbb{N}$  is also called  $\aleph_0$ , pronounced **Aleph-0**.
- A set  $X$  is called **uncountable** if it is infinite and not countable.
- The cardinality of  $\mathcal{P}(\mathbb{N})$  is written  $\mathfrak{c}$  and called the **continuum**.

REMARK 2.74. — A nontrivial fact is that

$$X \lesssim Y \quad \text{and} \quad Y \lesssim X \implies X \sim Y.$$

In other words, if there exist an injective map  $f : X \rightarrow Y$  and an injective map  $g : Y \rightarrow X$ , then one can find a bijective map  $h : X \rightarrow Y$ . This fact is known as the Schröder–Bernstein Theorem.

THEOREM 2.75 (Cantor’s diagonal argument). — *Let  $X$  be a set. Then the power set  $\mathcal{P}(X)$  is larger than  $X$  and not equal to  $X$ .*

*Proof.* The function  $i : X \rightarrow \mathcal{P}(X)$  given by  $i(x) = \{x\}$  is injective. So  $\mathcal{P}(X)$  is larger than  $X$ . It remains to show that there is no bijection from  $X$  to  $\mathcal{P}(X)$ . To show this, we assume that there is a bijection  $f : X \rightarrow \mathcal{P}(X)$  and take this to a contradiction. For this, we define the set

$$A = \{x \in X \mid x \notin f(x)\}.$$

In other words,  $A \in \mathcal{P}(X)$  consists of all elements  $x$  in  $X$  for which  $x$  is not an element of the subset  $f(x) \subset X$ .

Since, by assumption,  $f : X \rightarrow \mathcal{P}(X)$  is a bijection, there exists  $a \in X$  such that  $A = f(a)$ . We now ask ourselves: does  $a$  belong to  $A$  or not?

If  $a \in A$  then, by the definition of  $A$ ,  $a \notin f(a)$ . However, this is impossible since  $f(a) = A$ . Vice versa, if  $a \notin A$  it means that  $a \in f(a)$ , that again is impossible since  $f(a) = A$ .

This proves that there exists no  $a \in X$  with  $f(a) = A$ , which contradicts the surjectivity of  $f$ . So there can be no bijection  $f : X \rightarrow \mathcal{P}(X)$ .  $\square$

**PROPOSITION 2.76: UNCOUNTABILITY OF  $\mathbb{R}$**

*The set  $\mathbb{R}$  is uncountable.*

*Proof.* By Theorem 2.75,  $\mathcal{P}(\mathbb{N})$  is strictly larger than  $\mathbb{N}$ , and thus uncountable. Thus, to show that  $\mathbb{R}$  is uncountable, it suffices to prove the existence of an injection

$$\varphi : \mathcal{P}(\mathbb{N}) \rightarrow \mathbb{R}.$$

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We construct such an injection by assigning to each subset  $A$  the real number  $\varphi(A)$  whose decimal fraction expansion is given by  $a_0, a_1 a_2 a_3 a_4 \dots$  with

$$a_n = \begin{cases} 0 & \text{if } n \notin A \\ 1 & \text{if } n \in A \end{cases}$$

Injectivity of the function  $\varphi$  can be proved in two ways. As a first option, one can simply apply Problem 2.72. Alternatively, one can argue as follows. Let  $A$  and  $B$  be distinct subsets of  $\mathbb{N}$ , and let  $n$  be the smallest element of  $A \triangle B$  (recall Definition 2). If  $n \in A$  and  $n \notin B$  then  $\varphi(A) > \varphi(B)$  holds. On the other hand, if  $n \notin A$  and  $n \in B$  then  $\varphi(A) < \varphi(B)$  holds. Therefore  $\varphi(A) \neq \varphi(B)$  holds in all cases, which proves the injectivity.  $\square$

## 2.5 Sequences of Real Numbers

### 2.5.1 Convergence of Sequences

Let  $X$  be a set. Intuitively, a sequence in  $X$  is a non-terminating sequence of elements  $x_0, x_1, x_2, x_3, \dots$  of  $X$ . Here, we are interested in sequences and their properties in  $\mathbb{R}$ . We give now a precise definition, although we have already used the concept of sequence before in an intuitive way.

#### DEFINITION 2.77: SEQUENCES

A **sequence** in  $\mathbb{R}$  is a function  $a : \mathbb{N} \rightarrow \mathbb{R}$ . The image  $a(n)$  of  $n \in \mathbb{N}$  is also written as  $a_n$  and is called the  $n$ -th **element** of  $a$ . Instead of  $a : \mathbb{N} \rightarrow \mathbb{R}$  one often writes  $(a_n)_{n \in \mathbb{N}}$ ,  $(a_n)_{n=0}^{\infty}$ , or  $(a_n)_{n \geq 0}$ .

Since we primarily use the letter  $x$  to denote a real number, for sequences of real numbers we shall mostly use the notation  $(x_n)_{n \in \mathbb{N}}$ ,  $(x_n)_{n=0}^{\infty}$ , or  $(x_n)_{n \geq 0}$ .

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DEFINITION 2.78. — A sequence  $(x_n)_{n=0}^{\infty}$  is called **constant** if  $x_n = x_m$  for all  $m, n \in \mathbb{N}$ , and **eventually constant** if there exists  $N \in \mathbb{N}$  with  $x_n = x_m$  for all  $m, n \in \mathbb{N}$  with  $m, n \geq N$ .

EXAMPLE 2.79. — A sequence in  $\mathbb{R}$  is, for example,  $(x_n)_{n=0}^{\infty}$  given by  $x_0 = 1$ ,  $x_n = \frac{1}{n}$  for  $n \geq 1$ .

#### DEFINITION 2.80: CONVERGENCE FOR SEQUENCES

Let  $(x_n)_{n=0}^{\infty}$  be a sequence in  $\mathbb{R}$ . We say that  $(x_n)_{n=0}^{\infty}$  is **convergent** or **converges** if there exists an element  $A \in \mathbb{R}$  with the following property:

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall n \in \mathbb{N} : n \geq N \implies |x_n - A| < \varepsilon.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = A \tag{2.9}$$

and call the point  $A$  the **limit** of the sequence  $(x_n)_{n=0}^{\infty}$ .

It is a priori not clear that a converging sequence has only one limit. In the following lemma we show that the limit is indeed unique, so the notation (2.9) is justified.

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#### LEMMA 2.81: UNIQUENESS OF THE LIMIT

*A convergent sequence  $(x_n)_{n=0}^{\infty}$  in  $\mathbb{R}$  has exactly one limit.*



*Proof.* Let  $A \in \mathbb{R}$  and  $B \in \mathbb{R}$  be limits of the sequence  $(x_n)_{n=0}^\infty$ . Let  $\varepsilon > 0$ . Then we can find  $N_A, N_B \in \mathbb{N}$  such that for all  $n \geq N_A$  it holds  $|x_n - A| < \frac{\varepsilon}{2}$ , and for all  $n \geq N_B$  it holds  $|x_n - B| < \frac{\varepsilon}{2}$ . Set  $N = \max\{N_A, N_B\}$ . Then

$$|A - B| \leq |A - x_N| + |x_N - B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since  $\varepsilon > 0$  is arbitrary it follows that  $|A - B| = 0$ , and so  $A = B$ .  $\square$

EXAMPLE 2.82. — A constant sequence  $(x_n)_{n=0}^\infty$  with  $x_n = A \in \mathbb{R}$  for all  $n \in \mathbb{N}$  converges to  $A$ . Similarly, eventually constant sequences converge to the value they eventually take.

EXAMPLE 2.83. — The sequence of real numbers  $(\frac{1}{n})_{n=1}^\infty$  converges to zero, i.e.,  $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ . Indeed, given  $\varepsilon > 0$ , by Archimedes' principle (Theorem 2.63) there exists  $N \in \mathbb{N}$  with  $\frac{1}{N} < \varepsilon$ . Therefore, for every  $n \in \mathbb{N}$  with  $n \geq N$ , we have  $0 \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon$ .

EXAMPLE 2.84. — The sequence of real numbers  $(y_n)_{n=0}^\infty$  given by  $y_n = (-1)^n$  for  $n \in \mathbb{N}$  is not convergent, since the sequence members  $1, -1, 1, -1, 1, -1, \dots$  alternate between 1 and -1 and, in particular, do not approach any real number.

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## 2.5.2 Convergent Subsequences and Accumulation Points

Let  $(x_n)_{n=0}^\infty$  be a sequence in  $\mathbb{R}$ . We call a subsequence of  $(x_n)_{n=0}^\infty$  any sequence obtained by keeping only certain elements of the sequence  $(x_n)_{n=0}^\infty$  and ignoring all others. For example

$$x_0, x_1, x_4, x_9, x_{16}, x_{25}, \dots$$

is a subsequence. The formal definition is as follows:

### DEFINITION 2.85: SUBSEQUENCES

Let  $(x_n)_{n=0}^\infty$  be a sequence in  $\mathbb{R}$ . A **subsequence** of  $(x_n)_{n=0}^\infty$  is a sequence of the form  $(x_{n_k})_{k=0}^\infty$ , where  $(n_k)_{k=0}^\infty$  is a sequence of nonnegative integers such that  $n_{k+1} > n_k$  for all  $k \in \mathbb{N}$ .

REMARK 2.86. — In the previous definition, since  $n_{k+1} > n_k$  for all  $k \in \mathbb{N}$ , it follows that  $n_k \geq k$  for every  $k \in \mathbb{N}$ . (Exercise: prove this fact by induction on  $k \in \mathbb{N}$ .)

LEMMA 2.87. — Let  $(x_n)_{n=0}^\infty$  be a convergent sequence in  $\mathbb{R}$ . Each subsequence of  $(x_n)_{n=0}^\infty$  converges and has the same limit as  $(x_n)_{n=0}^\infty$ .

*Proof.* We leave the proof as an exercise.  $\square$

2.88. — A sequence can have convergent subsequences without itself converging. For example, the sequence of real numbers given by  $x_n = (-1)^n$  is not convergent, while the subsequences

$$(x_{2n})_{n=0}^{\infty} \quad \text{and} \quad (x_{2n+1})_{n=0}^{\infty}$$

are constant, and so in particular they are convergent.

#### DEFINITION 2.89: ACCUMULATION POINTS OF SEQUENCES

Let  $(x_n)_{n=0}^{\infty}$  be a sequence in  $\mathbb{R}$ . A point  $A \in \mathbb{R}$  is called an **accumulation point** of the sequence  $(x_n)_{n=0}^{\infty}$  if for every  $\varepsilon > 0$  and every  $N \in \mathbb{N}$  there exists a natural number  $n \geq N$  with  $|x_n - A| < \varepsilon$ .

#### PROPOSITION 2.90: SUBSEQUENCES AND ACCUMULATION POINTS

Let  $(x_n)_{n=0}^{\infty}$  be a sequence in  $\mathbb{R}$ . An element  $A \in \mathbb{R}$  is a accumulation point of  $(x_n)_{n=0}^{\infty}$  if and only if there exists a convergent subsequence of  $(x_n)_{n=0}^{\infty}$  with limit  $A$ .

*Proof.* Suppose  $A \in \mathbb{R}$  is an accumulation point of the given sequence. We recursively construct a subsequence  $(x_{n_k})_{k=0}^{\infty}$  that converges to  $A$ .

First, we apply the definition of accumulation point with  $\varepsilon = 1$  to find an integer  $n_0$  such that  $|x_{n_0} - A| \leq 1$  holds.

Then, we apply the definition of accumulation point with  $\varepsilon = 2^{-1}$  and  $N = n_0 + 1$  to find  $n_1 > n_0$  such that  $|x_{n_1} - A| \leq 2^{-1}$ .

Then, we apply the definition of accumulation point with  $\varepsilon = 2^{-2}$  and  $N = n_1 + 1$  to find  $n_2 > n_1$  such that  $|x_{n_2} - A| \leq 2^{-2}$ .

In general, we apply the definition of accumulation point with  $\varepsilon = 2^{-k}$  and  $N = n_{k-1} + 1$  to find  $n_k > n_{k-1}$  such that  $|x_{n_k} - A| \leq 2^{-k}$ .

We claim that the subsequence  $(x_{n_k})_{k=0}^{\infty}$  of  $(x_n)_{n=0}^{\infty}$  converges to  $A$ . Indeed, for every  $\varepsilon > 0$  there exists  $N \in \mathbb{N}$  with  $2^{-N} < \varepsilon$ . Then, for every  $k \geq N$  it holds

$$|x_{n_k} - A| \leq 2^{-k} \leq 2^{-N} < \varepsilon,$$

which shows that  $A$  is the limit of the sequence  $(x_{n_k})_{k=0}^{\infty}$ .

Conversely, suppose  $A$  is the limit of a converging subsequence  $(x_{n_k})_{k=0}^{\infty}$ . Fix  $\varepsilon > 0$  and  $N \in \mathbb{N}$ . Since  $x_{n_k}$  converges to  $A$ , there exists  $N_0 \in \mathbb{N}$  such that  $|x_{n_k} - A| < \varepsilon$  for all  $k \geq N_0$ . Hence, to satisfy the definition of an accumulation point, we want to choose  $k \geq N_0$  so that  $n_k \geq N$ . Recalling that  $n_k \geq k$  (see Remark 2.86), we define  $N_1 = \max\{N_0, N\}$  so that

$$|x_{n_{N_1}} - A| < \varepsilon \quad \text{and} \quad n_{N_1} \geq N.$$

So,  $A$  is an accumulation point of  $(x_n)_{n=0}^{\infty}$ . □

COROLLARY 2.91. — *A converging sequence has exactly one accumulation point, which coincides with its limit.*

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*Proof.* This follows from Lemma 2.87 and Proposition 2.90.  $\square$

EXERCISE 2.92. — Let  $(x_n)_{n=0}^{\infty}$  be a sequence in  $\mathbb{R}$ , and let  $F \subseteq \mathbb{R}$  be the set of accumulation points of the sequence  $(x_n)_{n=0}^{\infty}$ . Show that  $F$  is closed.

### 2.5.3 Addition, Multiplication, and Inequalities

2.93. — Given sequences  $(x_n)_{n=0}^\infty$  and  $(y_n)_{n=0}^\infty$ , and a number  $\alpha \in \mathbb{R}$ , sums and scalar multiples of sequences are given by

$$\begin{aligned}(x_n)_{n=0}^\infty + (y_n)_{n=0}^\infty &= (x_n + y_n)_{n=0}^\infty \\ \alpha \cdot (x_n)_{n=0}^\infty &= (\alpha x_n)_{n=0}^\infty.\end{aligned}$$

Sequences can also be multiplied as follows:

$$(x_n)_{n=0}^\infty \cdot (y_n)_{n=0}^\infty = (x_n y_n)_{n=0}^\infty.$$

REMARK 2.94. — With the addition and multiplication defined above, the set of sequences forms a commutative ring, where the zero element is the constant sequence  $(0)_{n=0}^\infty$ , and the neutral element for multiplication is the constant sequence  $(1)_{n=0}^\infty$ .

PROPOSITION 2.95. — Let  $(x_n)_{n=0}^\infty$  and  $(y_n)_{n=0}^\infty$  be sequences converging to  $A \in \mathbb{R}$  and  $B \in \mathbb{R}$ , respectively.

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1. The sequence  $(x_n + y_n)_{n=0}^\infty$  converges to  $A + B$ .
2. The sequence  $(x_n y_n)_{n=0}^\infty$  converges to  $AB$ , and the sequence  $(\alpha x_n)_{n=0}^\infty$  converges to  $\alpha A$  for every  $\alpha \in \mathbb{R}$ .
3. Suppose  $x_n \neq 0$  for all  $n \in \mathbb{N}$  and  $A \neq 0$ . Then the sequence  $(x_n^{-1})_{n=0}^\infty$  converges to  $A^{-1}$ .

*Proof.* 1. Let  $\varepsilon > 0$ , and let  $N_A, N_B \in \mathbb{N}$  be such that

$$|x_n - A| < \frac{\varepsilon}{2} \quad \forall n \geq N_A, \quad \text{and} \quad |y_n - B| < \frac{\varepsilon}{2} \quad \forall n \geq N_B.$$

Then, with  $N = \max\{N_A, N_B\}$ , for all  $n \geq N$  it holds

$$|(x_n + y_n) - (A + B)| \leq |x_n - A| + |y_n - B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which shows statement (1).

2. We refer to Exercise 2.96.

3. Let  $A$  be the limit of the sequence  $(x_n)_{n=0}^\infty$ . Since  $A \neq 0$ , choosing  $\varepsilon = \frac{|A|}{2}$  in the definition of limit, there exists  $N_0 \in \mathbb{N}$  such that, for all  $n \geq N_0$ ,  $|x_n - A| \leq \frac{|A|}{2}$ . In particular

$$|x_n| \geq \frac{|A|}{2} \quad \forall n \geq N_0,$$

which implies that

$$|x_n^{-1} - A^{-1}| = \frac{|x_n - A|}{|x_n| |A|} \leq 2 \frac{|x_n - A|}{|A|^2} \quad \forall n \geq N_0.$$

Now, given  $\varepsilon > 0$  we choose  $N_1$  so that  $|x_n - A| < \frac{|A|^2}{2}\varepsilon$  for all  $n \geq N_1$ . Then, for  $N = \max\{N_0, N_1\}$  we get

$$|x_n^{-1} - A^{-1}| \leq 2 \frac{|x_n - A|}{|A|^2} < \varepsilon \quad \forall n \geq N.$$

This proves that the sequence  $(x_n^{-1})_{n=0}^{\infty}$  converges to  $A^{-1}$ , as desired.  $\square$

EXERCISE 2.96. — Prove statement (2) in Proposition 2.95.